



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

ENEE4113 - COMMUNICATION LAB

Report#2

Experiment 6: Pulse Amplitude Modulation (Sampling)

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Abstract

This experiment aims to examine the applications of Pulse Amplitude Modulation (PAM) across various domains. It involves analyzing pulse train characteristics, studying modulation and demodulation techniques, investigating aliasing in both time and frequency domains, and exploring the concept of PAM multiplexing. Through this experiment, valuable insights into the significance of PAM in communication systems can be obtained.

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1.Theory

1.1 Part 1: Time and Frequency Characteristics of pulse train

Because the pulse train signal achieves infinite value in an infinitesimal length of time, it is impossible to recreate in the actual world. A popular method for performing sampling is to employ rectangular pulses with a period of T_s . This sampling strategy is known as sample-and-hold. Figure 1.1 depicts the sample-and-hold method of sampling:

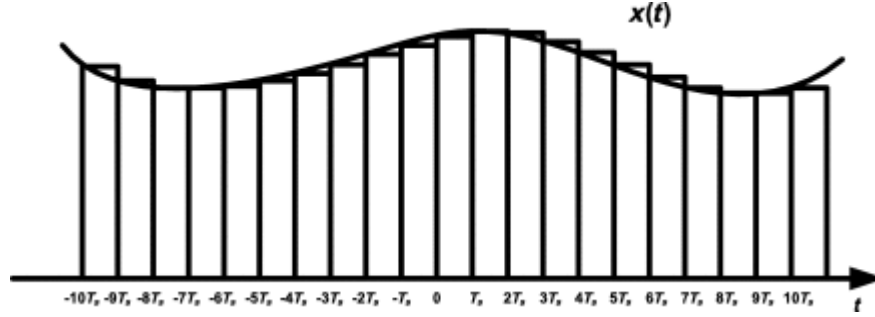


Figure 1.1: pulse train

The sample-and-hold approach can be described mathematically by the below equation, where the signal $p(t-nT_s)$ represents the periodic rectangular pulse.

$$X_{as}(t) = \sum_{n=-\infty}^{\infty} x(nT_s) p(t - nT_s)$$

The periodic rectangular pulse can be rewritten as below:

$$\sum_{n=-\infty}^{\infty} p(t - nT_s) = p(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Substituting these two equations, it's got:

$$x_{as}(t) = p(t) * \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s)$$

The right term of the equation can be rewritten as shown:

$$p(t) * \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) = p(t) * (x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s))$$

Eq. (1) shows the Fourier transform of $X_{as}(t)$

$$X_{as}(\omega) = \frac{\sin\left(\frac{\omega T_s}{2}\right)}{\omega} e^{-\frac{j\omega T_s}{2}} \cdot \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(\omega - k\omega_s) \quad \text{Eq. (1)}$$

Eq. (2) shows a simplification of Eq. (1):

$$X_{as}(\omega) = \frac{\sin\left(\frac{\omega T_s}{2}\right)}{\omega} e^{-j\frac{\omega T_s}{2}} X_a(\omega) \quad \text{Eq. (2)}$$

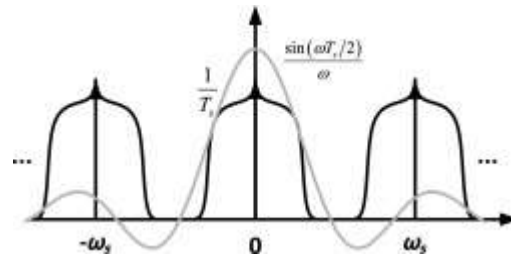
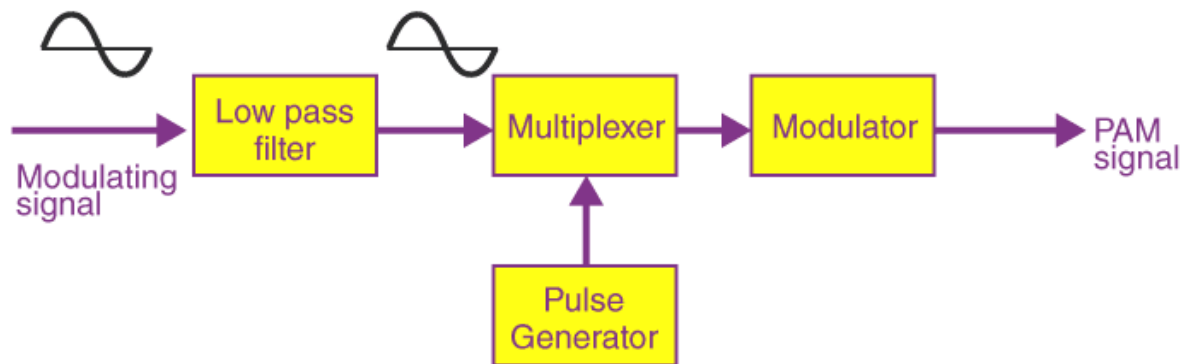


Figure 1.1: Spectrum obtained by the ideal sampling using sample-and-hold

Part 2: Characteristics of Pulse Amplitude Modulation (PAM)

The data transmission method of pulse amplitude modulation is defined as changing the amplitudes (power levels or voltage) of each pulse in a regular temporal sequence of electromagnetic pulses. The number of possible amplitudes is limitless, although it is usually some power of two so that the final output signal can be digital. Level-4 PAM, for example, has 22 distinct pulse amplitudes; level-8 PAM has 23 discrete pulse amplitudes.



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Figure 1.2: Pulse Amplitude Modulation Block Diagram

The amplitudes of regularly spaced rectangular pulses fluctuate one-to-one with the instantaneous sample values of a continuous message stream in pulse-amplitude modulation (PAM). This modulation approach is depicted in Figures 6a, b, and c, which depict a message signal, a pulse carrier, and the matching PAM wave, respectively.

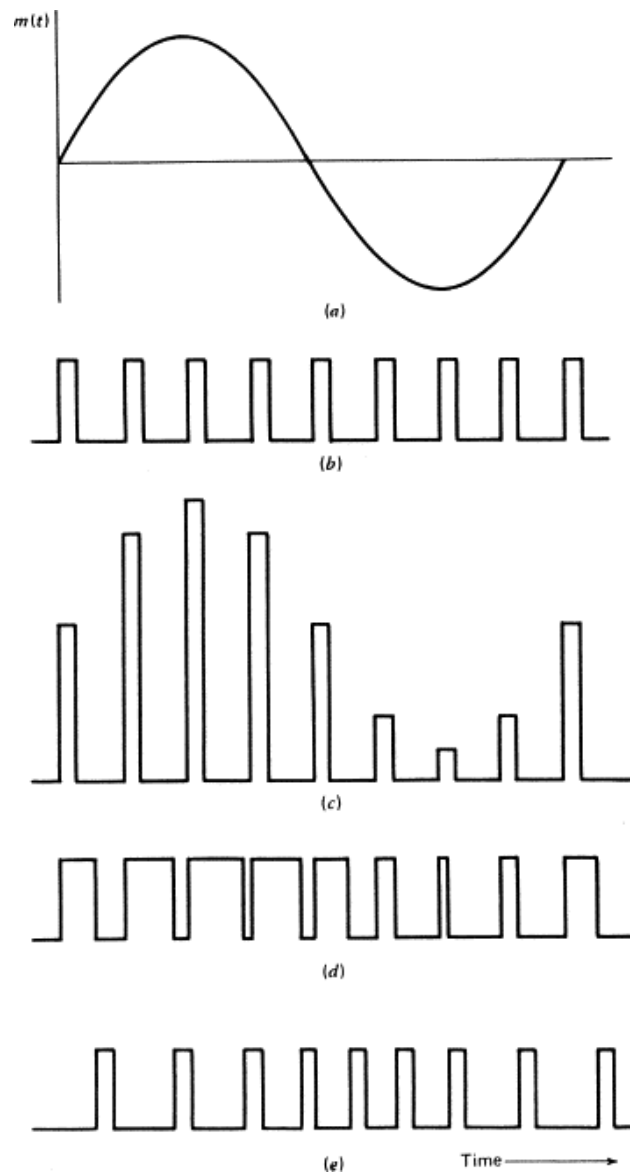


Figure 1.2: Different forms of pulse-analog modulation for the case of a sinusoidal modulating wave

A PAM system can be viewed as a switch that closes every T_s seconds for seconds, but otherwise remains open. The PAM signal is therefore the result of multiplying the continuous-time signal $x(t)$ by a periodic signal $p(t)$ composed of pulses of width, amplitude $1/\Delta$, and period T_s . Thus, $X_{PAM}(t)$ is made up of narrow pulses with signal amplitudes within the pulse width (for narrow pulses, the pulse amplitude can be approximated by the signal amplitude at the sampling time (this is known as flat-top PAM)).

Part 3: Characteristics of Pulse Amplitude Modulation

Demodulation recovers the original modulating frequencies from the carrier signal using a detector circuit. It aims to faithfully replicate the modulation to avoid distortion. Demodulation methods vary based on how the baseband signal characteristics are transmitted. Nonlinear devices are used for extracting modulating frequencies from the envelope.

A demodulator circuit that removes the modulating signal from a Single Polarity PAM pulse is demonstrated. A Wien Bridge Oscillator (WBO) generates the modulating signal, and PAM modulation is accomplished via a FET-based modulator circuit. PAM signal demodulation is simple and may be achieved with a low-pass filter.

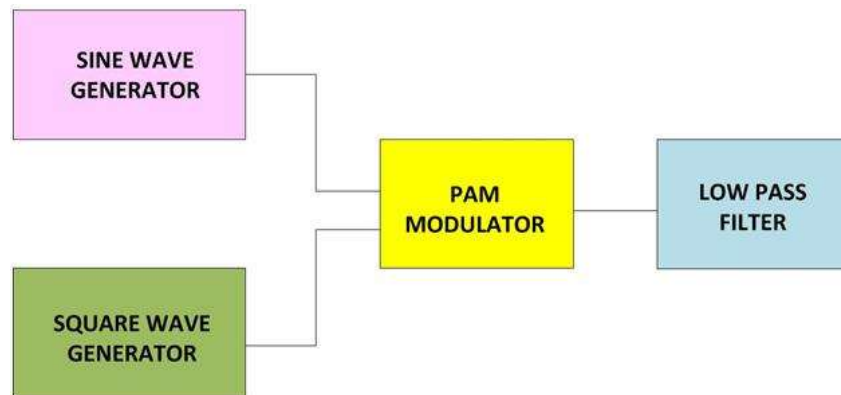


Figure 1.3: Block Diagram Of Pulse Amplitude Modulation (PAM)

Part 4: Aliasing in the Time and the Frequency Domains

Aliasing is the effect of new frequencies appearing in the sampled signal after reconstruction, that were not present in the original signal. It is caused by too low sample rate for sampling a particular signal or too high frequencies present in the signal for a particular sample rate. We can avoid it by using sufficiently large sample rate or low-pass filtering the signal before sampling to ensure the $f_{max} < \frac{f_s}{2}$ condition.

Now with aliasing $f_{max} < \frac{f_s}{2}$, so we could observe something like this:

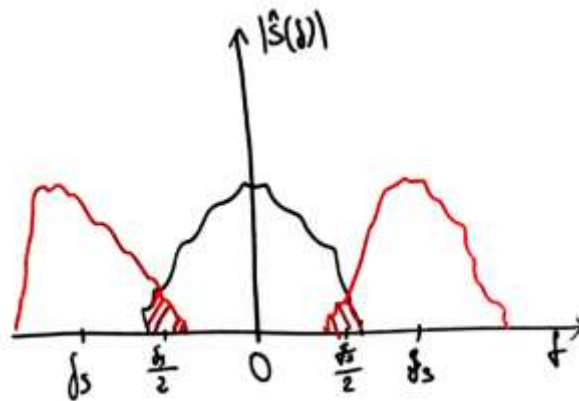


Figure 1.4: Aliased spectra

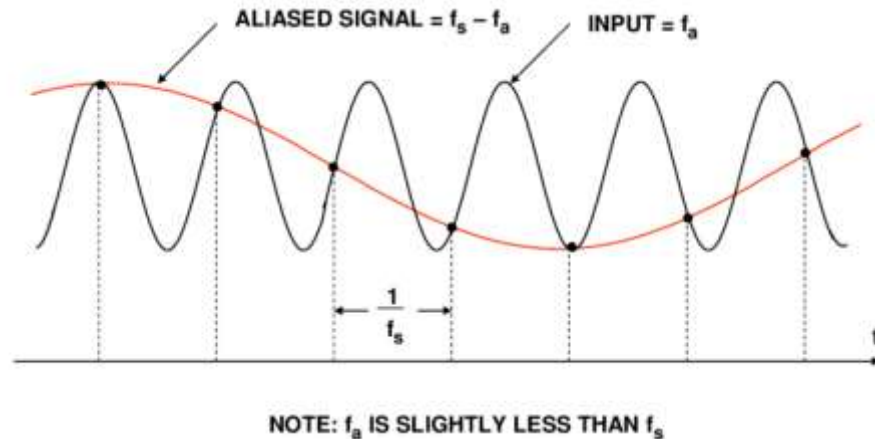


Figure 1.4: Aliasing in the Time Domain

Part 5: PAM Time Multiplex

Multiplexing, as we know, permits the transmission of several signals over a single channel. However, for appropriate data transfer, it may be necessary to distinguish between the various signals. Thus, time division multiplexing transmits the entire signal by occupying distinct time slots.

The word itself indicates that time division is used in order to multiplex numerous data channels. The figure below shows the block diagram of a TDM system employing both transmitter and receiver section.

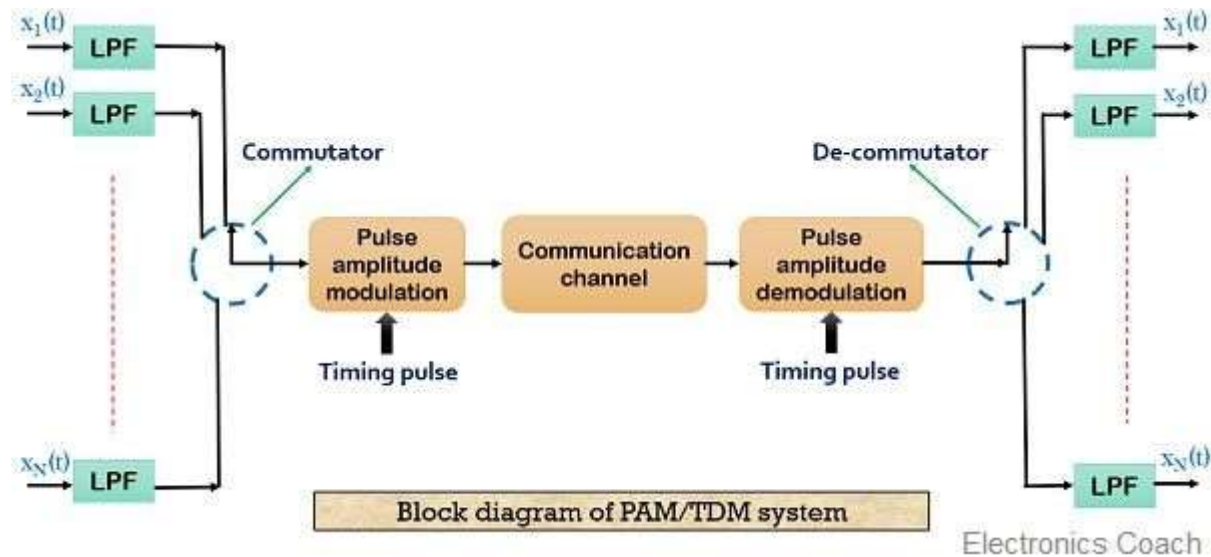


Figure 1.5: Block diagram of PAM/TDM system

Because the approach efficiently uses the entire channel for data transmission, it is also known as PAM/TDM. This is due to the fact that a TDM system employs pulse amplitude modulation. Each pulse in this modulation method has a short time length, allowing for maximum channel use.

2. Procedure

Part 1: Time and Frequency Characteristics of pulse train

The components were assembled as shown below in figure 2.1.



Figure 2.1: Pulse train components

1- The pulse train was selected at the function generator with:

Freq = 1 kHz, $V_{ss} = 10$ V and duty cycle = 10%.

2- The measurement was started.

3- The pulse train was showed in the time and frequency domains.

4- The measurement of the spectral and time characteristics was repeated for the same frequency and amplitude for different duty cycles 20%, 30%, 40, 50% and 90%.

Part 2: Characteristics of Pulse Amplitude Modulation (PAM)

The components were assembled as shown below in figure 2.2. (Sync Pin was neglected)

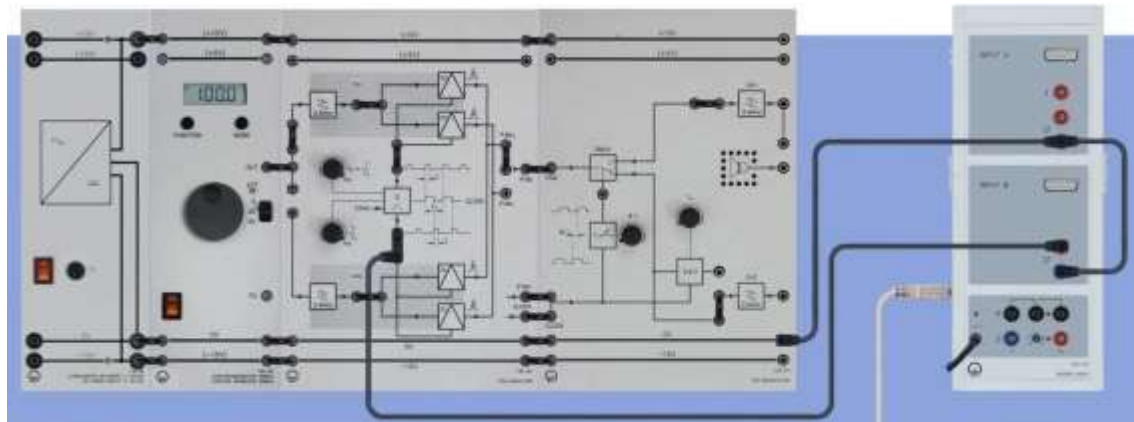


Figure 2.2: PAM modulation components

1- The sampling frequency (f_p) knob was adjusted to its max value.

2- The duty cycle (τ/T_P) knob was adjusted to its max value.

3- The Cassy UA1 was connected to the Clock generator output (G).

4- The sampling frequency was adjusted to 5000 Hz.

After adjusting the sampling frequency, the Time and the Frequency characteristics was investigated of the Pulse Amplitude Modulation.

1.Study the effect of the channel filter (CH1):

To study the effect of any filter, a signal of a specified amplitude and frequency is passed through the filter. Then the output signal of the filter is compared with the original one.

- 1- The function generator was set to:

Sine wave, Freq = 500 Hz and Vss = 10V

- 2- The Cassy sensor UA1 was connected to the input message signal.
- 3- The Cassy sensor UB1 was connected to the output signal of the CH1 filter.
- 4- The measurement was started.

2.Display the Pulse Amplitude Modulated signal s(t) of PAM1 in the time domain:

1. The function generator was set to:

Sine wave, Freq = 500 Hz and Vss =

- 10V
2. The Cassy sensor UA1 was connected to the output signal of the CH1 filter.
3. The Cassy sensor UB1 was connected to the PAM1 output.
4. The measurement was started.

2.1 Determining the effect of the duty cycle on the PAM in the time domain:

5. The duty cycle of the clock generator was reduced to the min value.
6. The measurement was started.

3. Display the Pulse Amplitude Modulated signal s(t) of PAM1 in the Frequency domain:

The same connection that was used to study the time domain of PAM1 was used.

1. The duty cycle of the clock generator was set to the max value.
2. The Cassy UA1 was connected to the Clock generator output.
3. The PAM1 output was showed in the frequency domain.

3.1 Determining the effect of the message frequency on the PAM in the Freq domain:

4. The message frequency was changed to 1kHz and 2kHz respectively.

3.2 Determining the effect of the duty cycle on the PAM in the Freq domain:

5. The message frequency was set back to 500Hz.
6. The duty cycle of the clock generator was changed to 10% and 30%.

7. After finishing this part the duty cycle was returned to 50%.

4. Display the Pulse Amplitude Modulated signal $s(t)$ of PAM2 in the time domain:

1. The function generator was set to:
Sine wave, Freq = 500 Hz and Vss = 10V
2. The Cassy sensor UA1 was connected to the output signal of the CH1 filter.
3. The Cassy sensor UB1 was connected to the PAM2 output.
4. The measurement was started.

4.1 Determining the effect of the duty cycle on the PAM in the time domain:

5. The duty cycle of the clock generator was reduced to the min value.
6. The measurement was started.

5. Display the Pulse Amplitude Modulated signal $s(t)$ of PAM2 in the Frequency domain:

The same connection that was used to study the time domain of PAM2 was used.

1. The duty cycle of the clock generator was set to the max value.
2. The Cassy UA1 was connected to the Clock generator output.
3. The PAM2 output was showed in the frequency domain.

5.1 Determining the effect of the message frequency on the PAM in the Freq domain:

4. The message frequency was changed to 1kHz and 2kHz respectively.

5.2 Determining the effect of the duty cycle on the PAM in the Freq domain:

5. The message frequency was set back to 500Hz.
6. The duty cycle of the clock generator was changed to 10% and 30%.
7. After finishing this part the duty cycle was returned to 50%.

Part 3: Characteristics of Pulse Amplitude Demodulation

a) Display the message signal and the demodulated signal using PAM1:

1. The clock generator frequency was set to its max value.
2. The function generator was set to:
Sine wave, Freq = 500 Hz and Vss = 10V
3. The Cassy sensor UA1 was connected to the input signal of the CH1 filter.
4. The Cassy sensor UB1 was connected to the output of the demodulator filter of CH1.
5. The measurement was started.

b) Determining the effect of the duty cycle on the PAM in the time domain:

6. The duty cycle of the clock generator was changed to 10% and 30% respectively.

Part 4: Aliasing in the Time and the Frequency Domains

a) **Display the message signal and the demodulated signal using PAM1 in the time domain:**

1. The clock generator frequency was set to 5000 Hz.
2. The duty cycle was set to 50%.
3. The function generator was set to:
Sine wave, Freq = 3000 Hz and $V_{ss} = 5V$
4. The Cassy sensor UA1 was connected to the input signal of the CH1 filter.
5. The Cassy sensor UB1 was connected to the PAM1 modulator.
6. The measurement was started.
7. The Cassy sensor UB1 was connected to the output of the demodulator filter of CH1.
8. The measurement was started.

b) **Display the message and the demodulated signals using PAM1 in the frequency domain:**

The same connection that was used to study the time domain of PAM1 was used.

- 1- The Cassy sensor UA1 was connected to the output of the clock generator.
- 2- The Cassy sensor UB1 to the PAM1 modulator.
- 3- FFT measurement was started.
- 4- The Cassy sensor UB1 was connected to the output of the demodulator filter of CH1.
- 5- The measurement was started.

Part 5: PAM Time Multiplex

The components were assembled as shown below in figure 2.2. (Sync Pin was neglected)

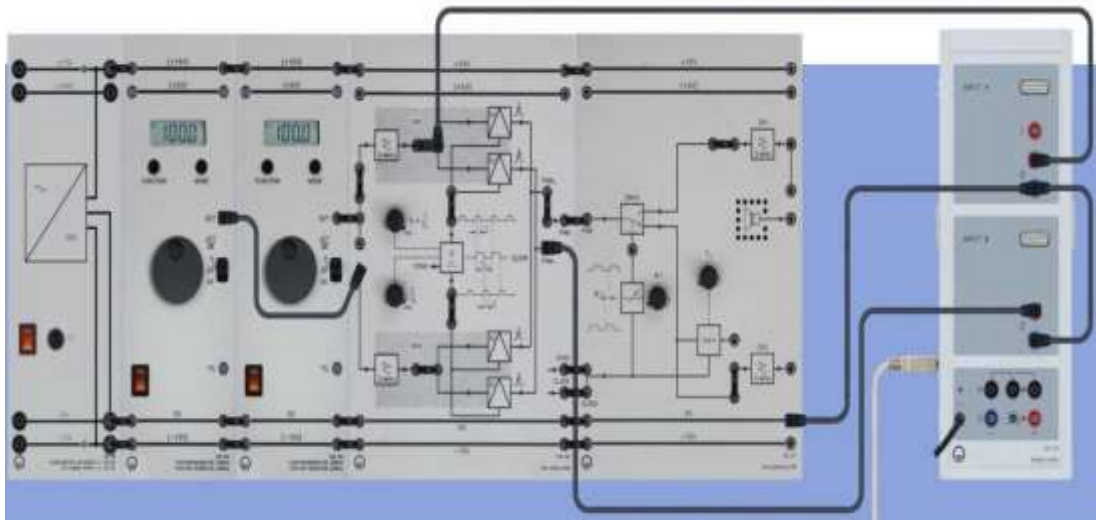


Figure 2.5: PAM time Multiplex components

a) Display the time characteristic of the time multiplex signal.

1. The sampling frequency f_P was set to the max. And the duty cycle was set to its max value.
2. The first function generator was set to:
Triangle, $f_{M1} = 200 \text{ Hz}$, $V_{SS} = 5 \text{ V}$
3. The second function generator was set to:
Sine, $f_{M2} = 300 \text{ Hz}$, $V_{SS} = 10 \text{ V}$
4. The CASSY sensor UA1 was connected to the triangle message channel CH1.
5. The CASSY sensor UB1 was connected to Output PAM modulator PAM1.
6. The measurement was started by pressing F9.
7. The CASSY sensor UA1 was connected to the sine message channel CH1.
8. The measurement was started by pressing F9.
9. The CASSY sensor UA1 was connected to the demodulator filter CH1.
10. The CASSY sensor UB1 was connected to the demodulator filter CH2.

b) Cross Talk (Δt) left/middle

1. The same connections of the message signals was kept.
2. The effect of cross talk as the phase shift Δt changes in both the time and the frequency domains was observed.
3. The first function generator was set to:
Triangle, $f_{M1} = 200 \text{ Hz}$, $V_{SS} = 0 \text{ V}$
4. The second function generator was set to:
Sine, $f_{M2} = 300 \text{ Hz}$, $V_{SS} = 10 \text{ V}$
5. The effect of cross talk as the phase shift Δt changes in both the time and the frequency domains was observed.

3.Data & Results

Part 1: Time and Frequency Characteristics of pulse train

After sitting the required information, the pulse train in the time domain it appeared as in the figure below:

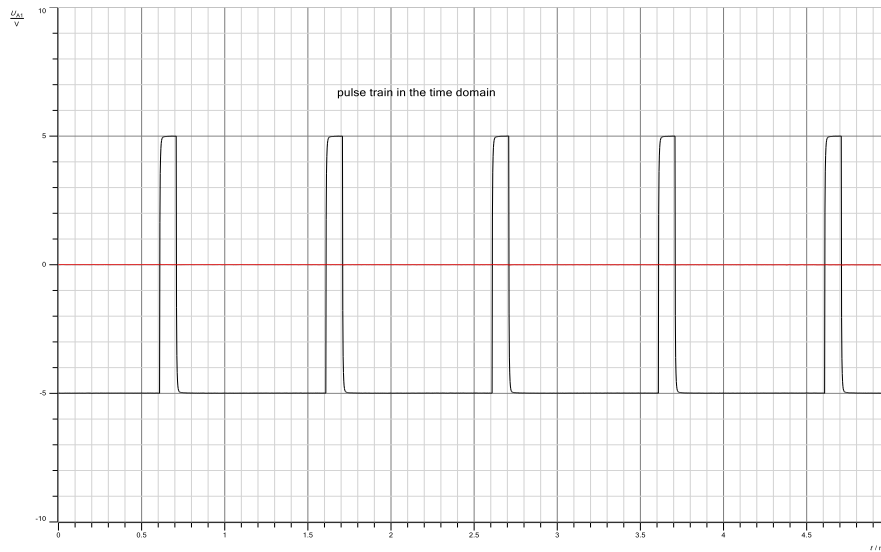


Figure 3.1: pulse train at time domain with duty cycle 10%

The signal exhibited a fleeting surge, reaching a voltage of 5 volts, lasting only for 0.1 milliseconds within a 1-millisecond timeframe (equivalent to a 10% duty cycle). Owing to the inherent imperfections of the generator, a slight lag occurs during the signal's transition from 0 volts to 5 volts and vice versa.

And in the frequency domain it appeared as in the figure below:

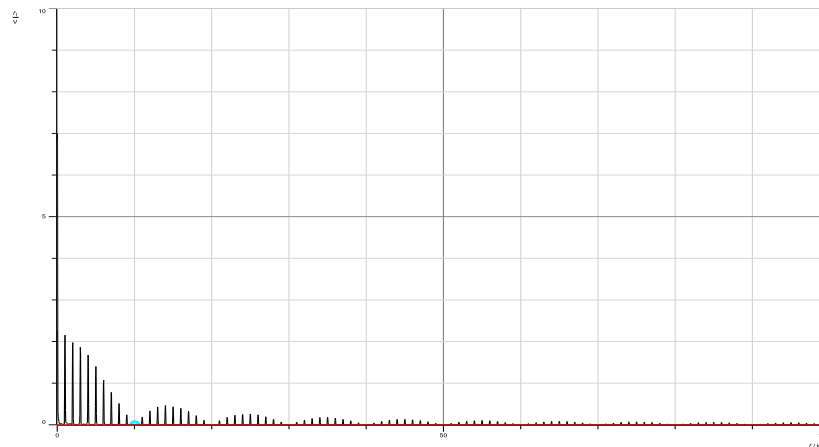


Figure 3.1: pulse train at frequency domain with duty cycle 10%

- The zero crossings in the envelope of the pulse spectrum was at 10 as the output.

The impulse values become zero for frequencies of 10 kHz and higher. These zero-crossing points align with the formula for calculating Fourier series coefficients:

$$X_n = \tau / T_0 \operatorname{sinc}\left(n \left(\frac{\tau}{T_0}\right)\right)$$

The *sinc* function is zero when $n (\tau/T_0)$ is an integer value, hence:

$n (\tau/T_0) = k$, where k is some integer

$n (1/10) = k \Rightarrow n = 10m$ so that k is an integer

As a result, 9 spectrum lines appear between each pair of zero crossings.

Then the measurement is repeated of the **spectral** and **time** characteristics for the same frequency and amplitude for different duty cycles 20%, 30%, 40, 50% and 90%, there is the output at duty cycle 20%:

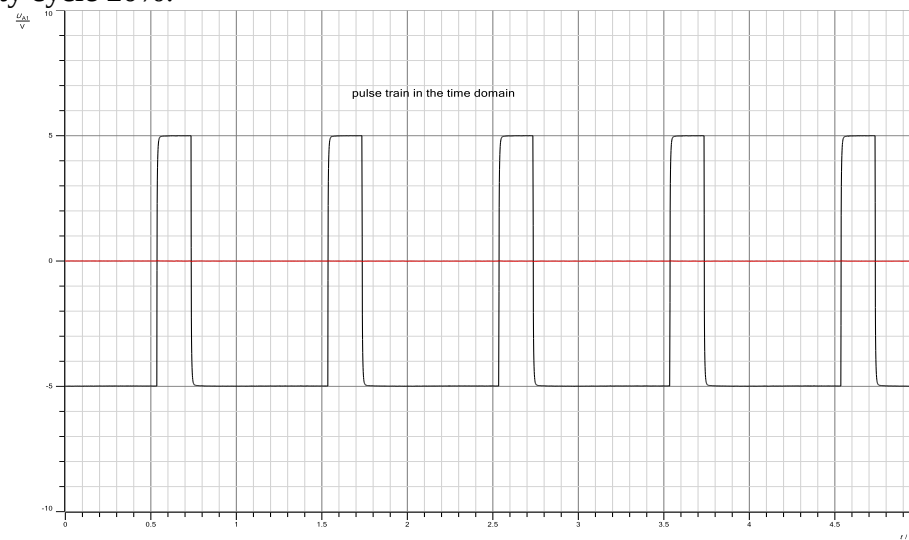


Figure 3.1: pulse train at time domain with duty cycle 20%

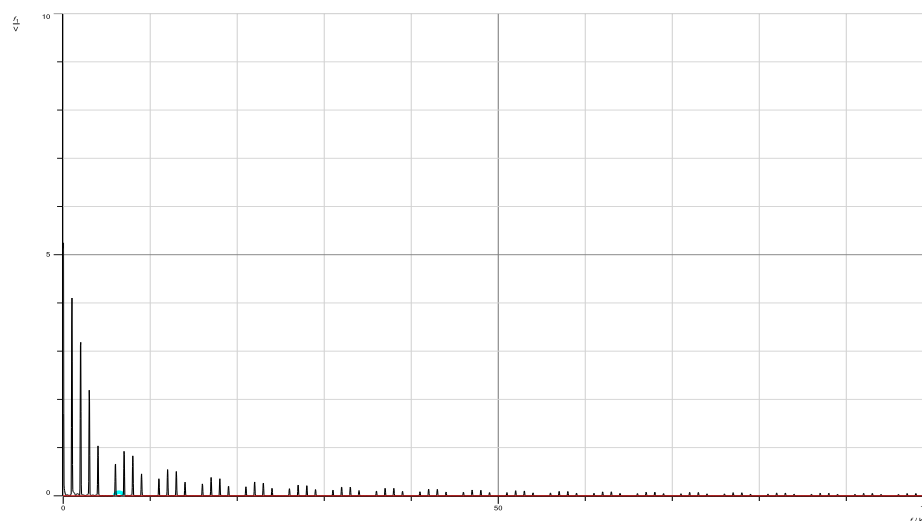


Figure 3.1: pulse train at frequency domain with duty cycle 20%

The zero-crossing points are at multiples of 5 kHz. This matches the theoretical value:
 $n(2/10) = k \Rightarrow n = 5m$ so that k is an integer

And The table below is resulted.

Table 3.1: Pulse train table

Duty Cycle (%)	10	20	30	40	50	90
Zero Crossing (kHz)	10	4.64	3.33	2.5	2	1.11

- Due to the quick changes between high and low amplitudes, pulse trains necessitate wide transmission bandwidths. Because of the quick fluctuations in the signal, a wider bandwidth is required for reliable transmission and correct replication of the pulse train.
- A pulse train has a spectrum that is made up of discrete frequency components that are usually harmonics of the fundamental frequency. The pulse train's envelope represents amplitude fluctuations over time, offering an overview of the signal's intensity changes.

Part 2: Characteristics of Pulse Amplitude Modulation (PAM)

When entering the data that was needed, the Characteristics of Pulse Amplitude Modulation (PAM) showed up as the follow:

The pulse frequency (f_p) was reduced, until the spectral line of the fundamental pulse appears at ($f_0 = 5000$ Hz) and appeared in the following way:

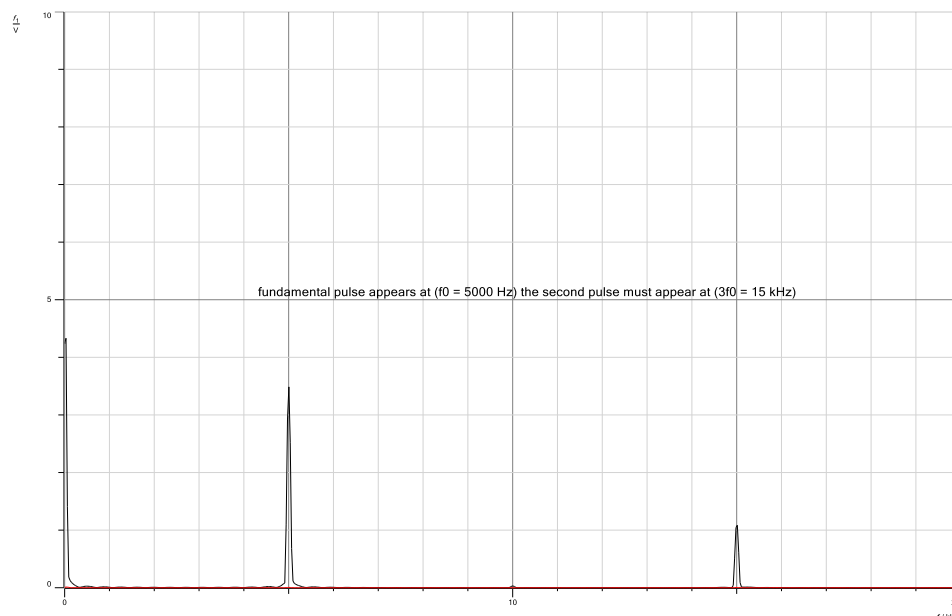


Figure 3.2: fundamental pulse appears at 5kHz

a) Study the effect of the channel filter (CH1):

A signal of a specific amplitude and frequency passes through a filter to analyze its impact. The filter's output signal is then compared to the original. And this is the outcome as a result of it:

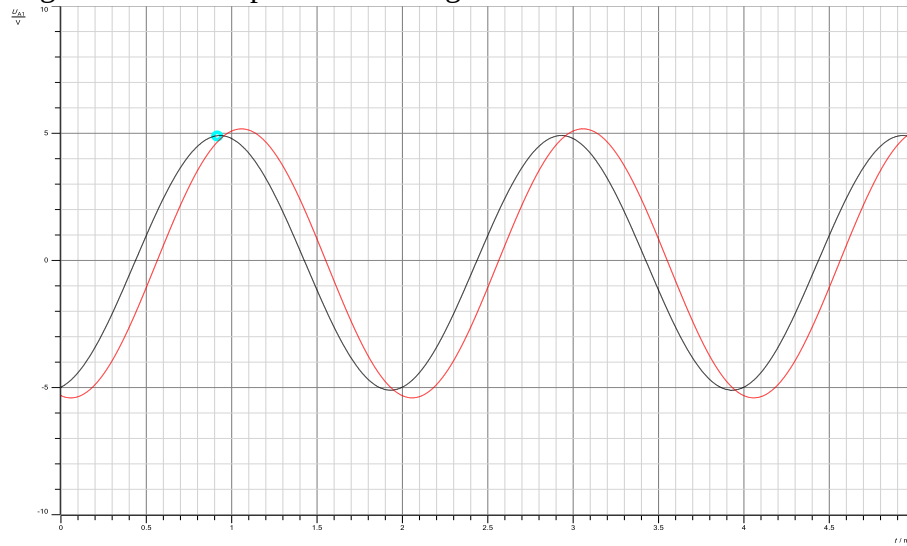


Figure 3.2: the effect of the filter on PAM

- The gain of the filter at the 500 Hz as shown by experience is $5.18/4.9 = 1.057$

PAM's channel filter, with a gain of 1.057, modifies the spectrum of the transmitted signal. It selectively transmits desirable frequencies while attenuating others, so improving signal quality and decreasing interference. As a result, communication is clearer and more trustworthy, with less distortion.

b) Display the Pulse Amplitude Modulated signal $s(t)$ of PAM1 in the time domain:

After change the connection as required, this is the output: (At max duty cycle)

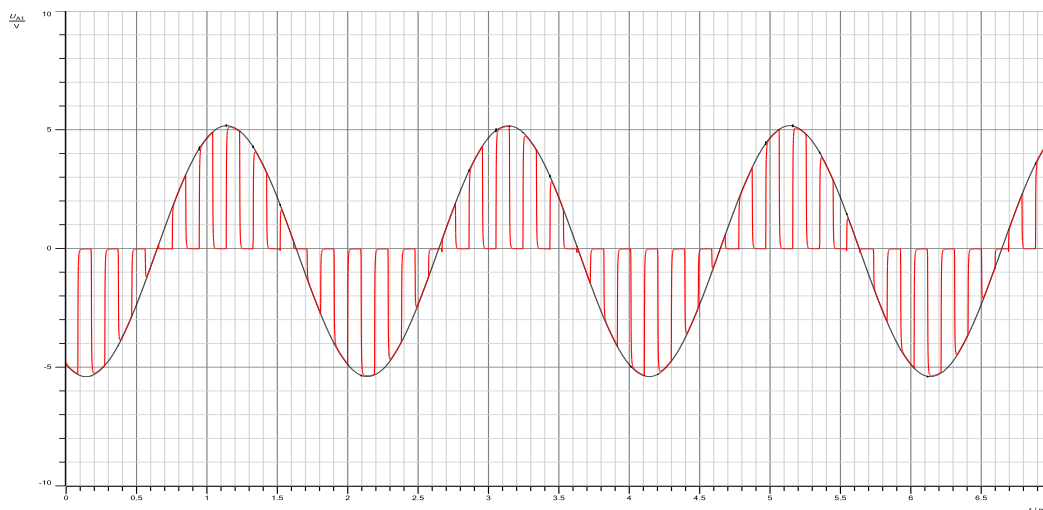


Figure 3.2: The Pulse Amplitude Modulated signal of PAM1 in the time domain

- **determining the effect of the duty cycle on the PAM in the time domain:**

The duty cycle of the clock generator was reduced to the min value.

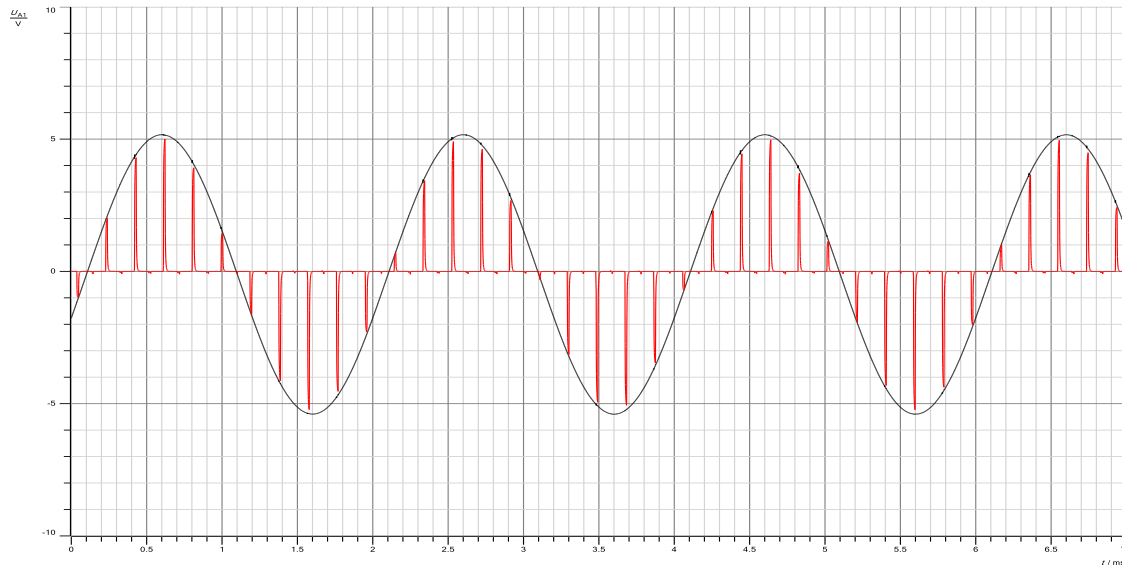


Figure 3.2: $s(t)$ at the min duty cycle

The duty cycle of PAM affects the pulse length and transition period. Longer pulses arise from higher duty cycles, whereas shorter pulses result from lower duty cycles.

c) **Display the Pulse Amplitude Modulated signal $s(t)$ of PAM1 in the Frequency domain:**

After use the same connection that was used to study the time domain of PAM1, the duty cycle was set to the max value and this is the output in the frequency domain:

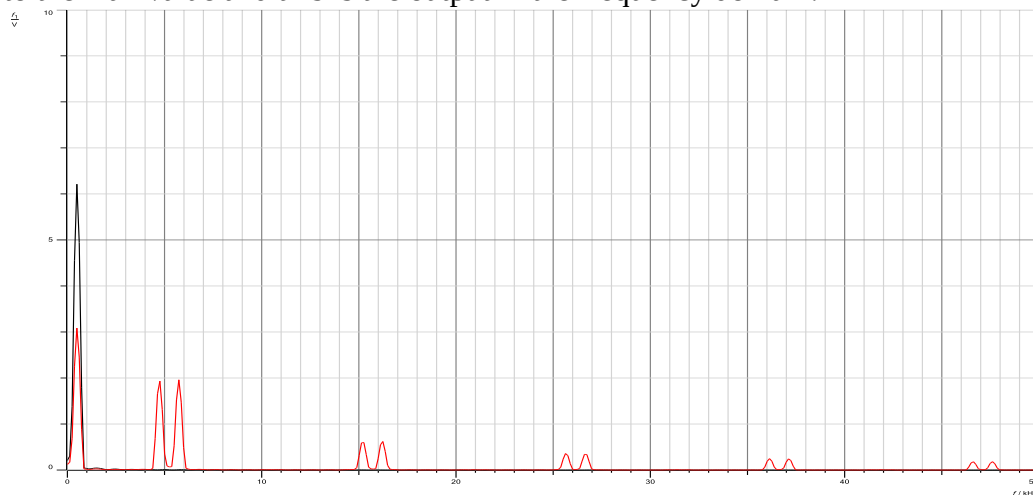


Figure 3.2: $s(t)$ of PAM1 in the frequency domain

In the frequency domain, the signal has an obvious spectral component at the carrier frequency, which indicates the modulating signal frequency, such as 5 kHz or 15 kHz.

- **Determining the effect of the message frequency on the PAM in the Freq domain:**

Then the message frequency was changed to 1kHz and the output shown as the below:

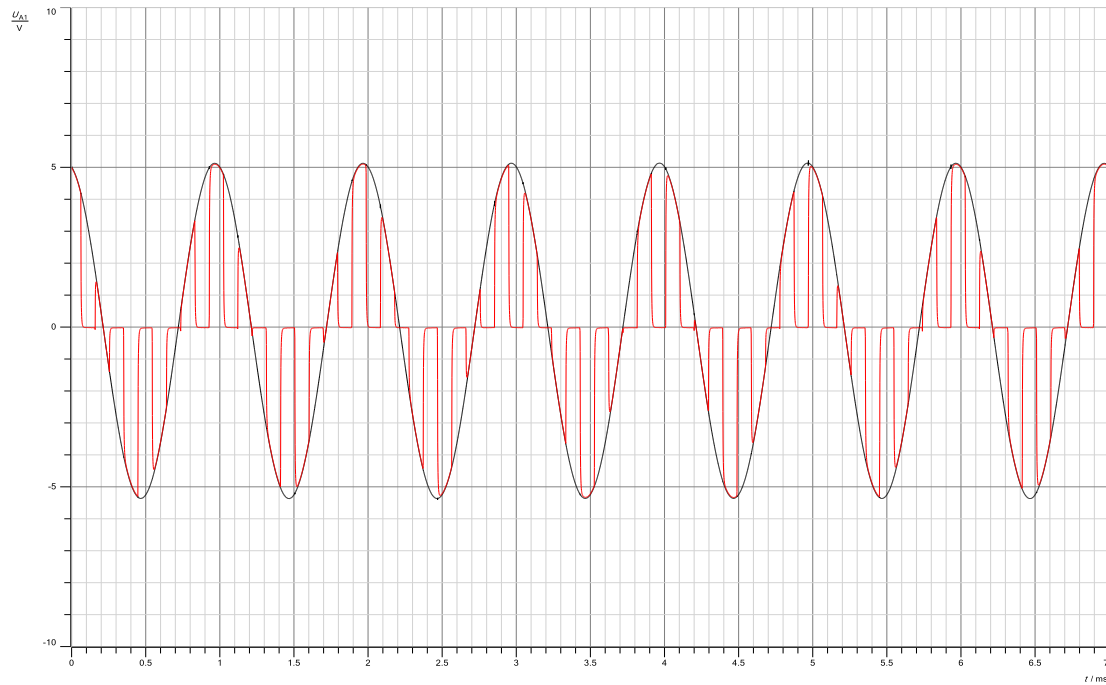


Figure 3.2: PAM signal at 1kHz

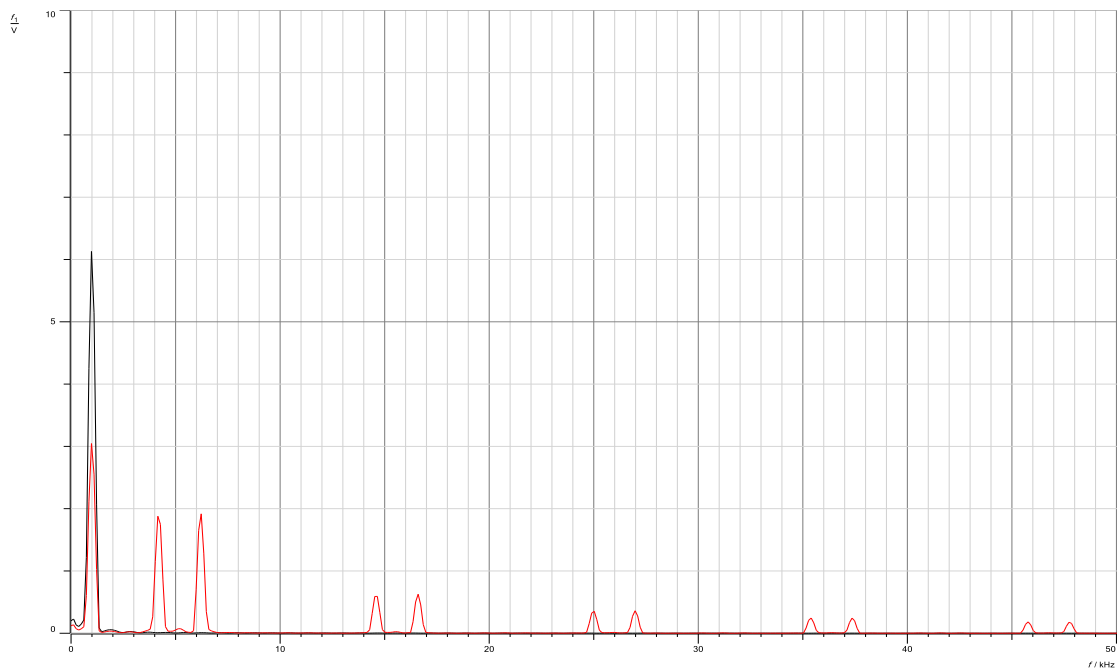


Figure 3.2: PAM signal at 1kHz in frequency domain

And then at 2kHz:

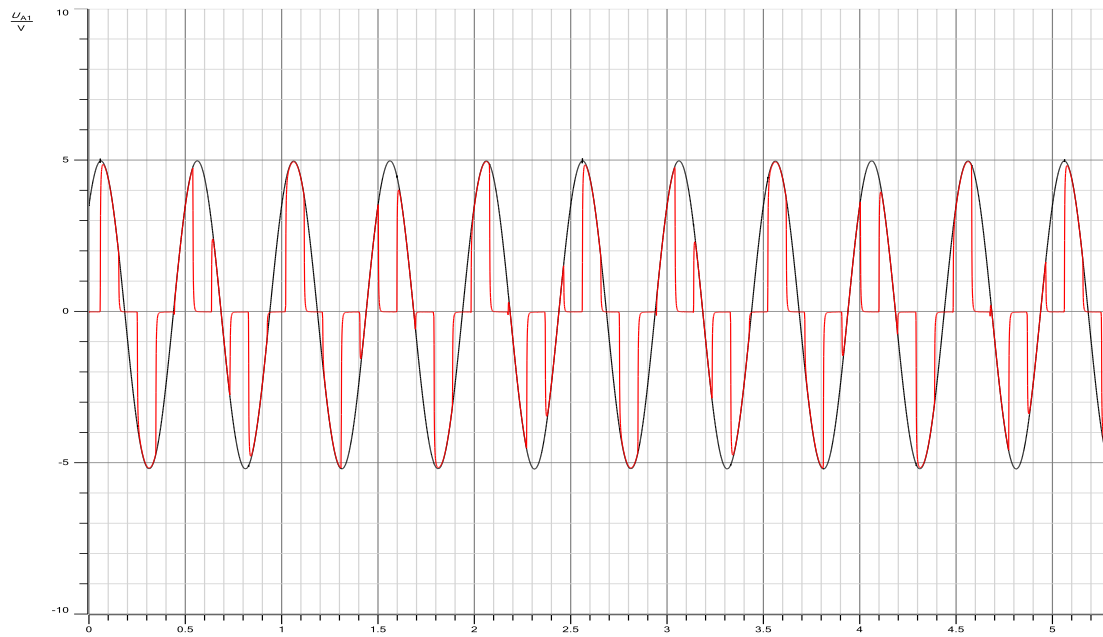


Figure 3.2: PAM signal at 2kHz

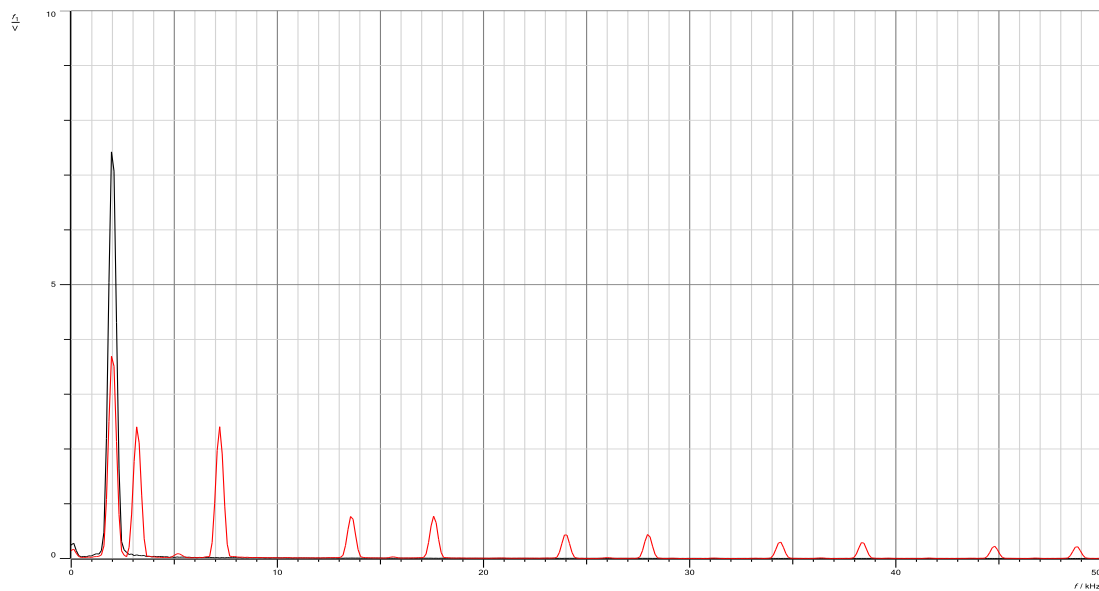


Figure 3.2: PAM signal at 2kHz in frequency domain

The message frequency has an influence on both the frequency and time domains. Increasing the message frequency, for example, from 1 kHz to 2 kHz, expands the frequency domain spectrum, resulting in closely spaced frequency components. Higher message frequencies result in shorter pulse durations and a higher rate of pulse repetition in the time domain.

- **Determining the effect of the duty cycle on the PAM in the Freq domain:**

After set the message frequency back to 500Hz, the duty cycle was changed to 10% and this the output:

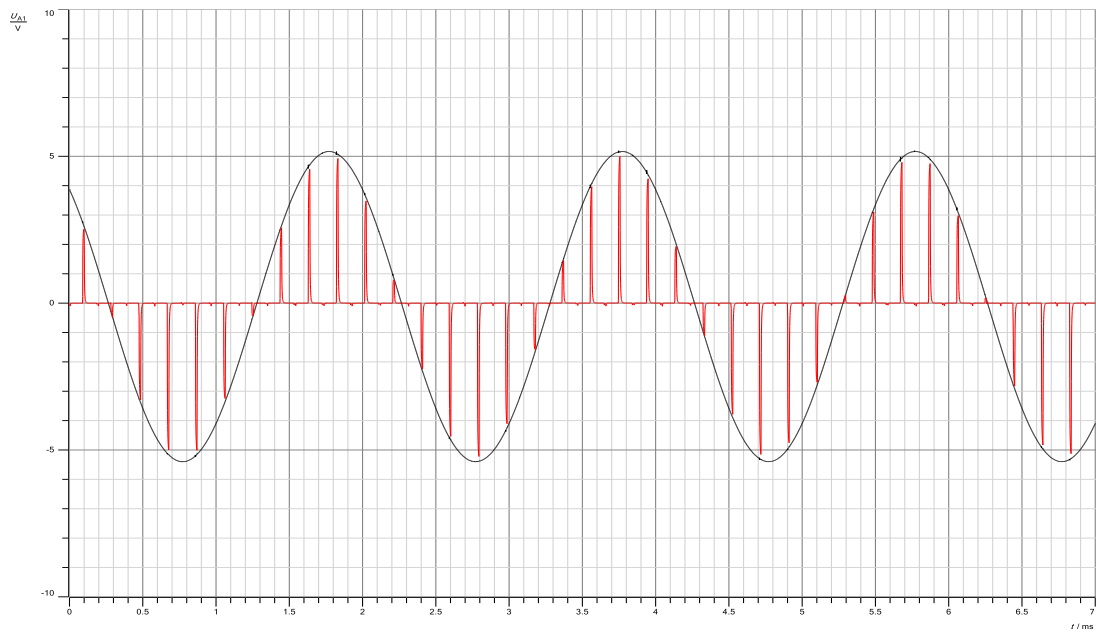


Figure 3.2: the signal at 500Hz & duty cycle 10%

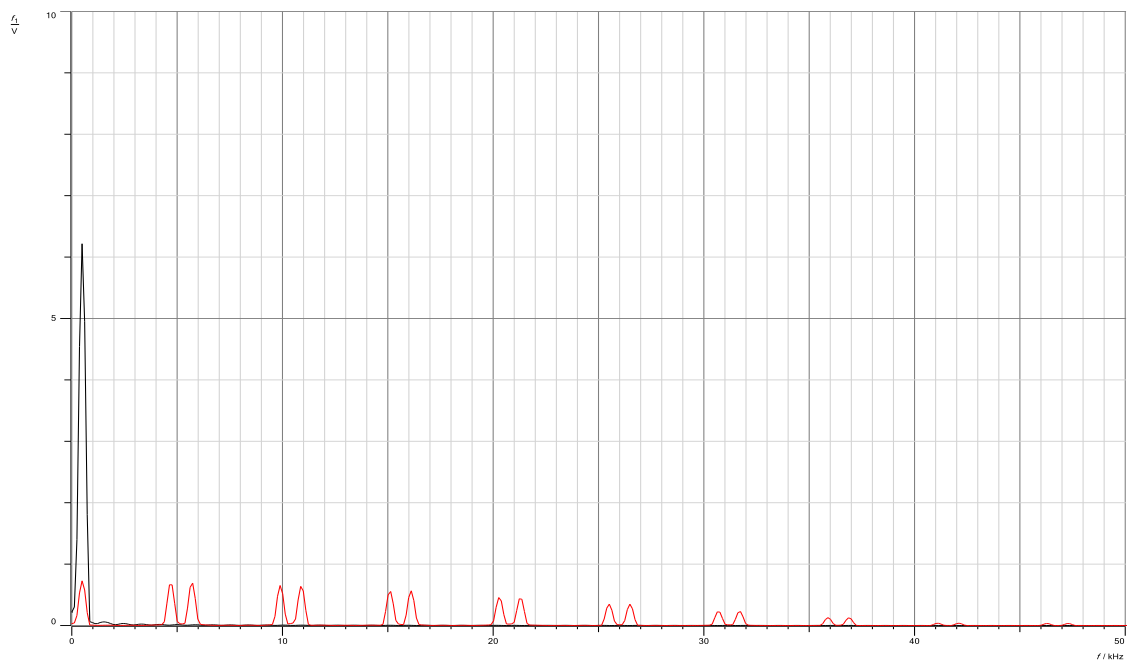


Figure 3.2: the signal at 500Hz & duty cycle 10% in frequency domain

And then at 30% duty cycle:

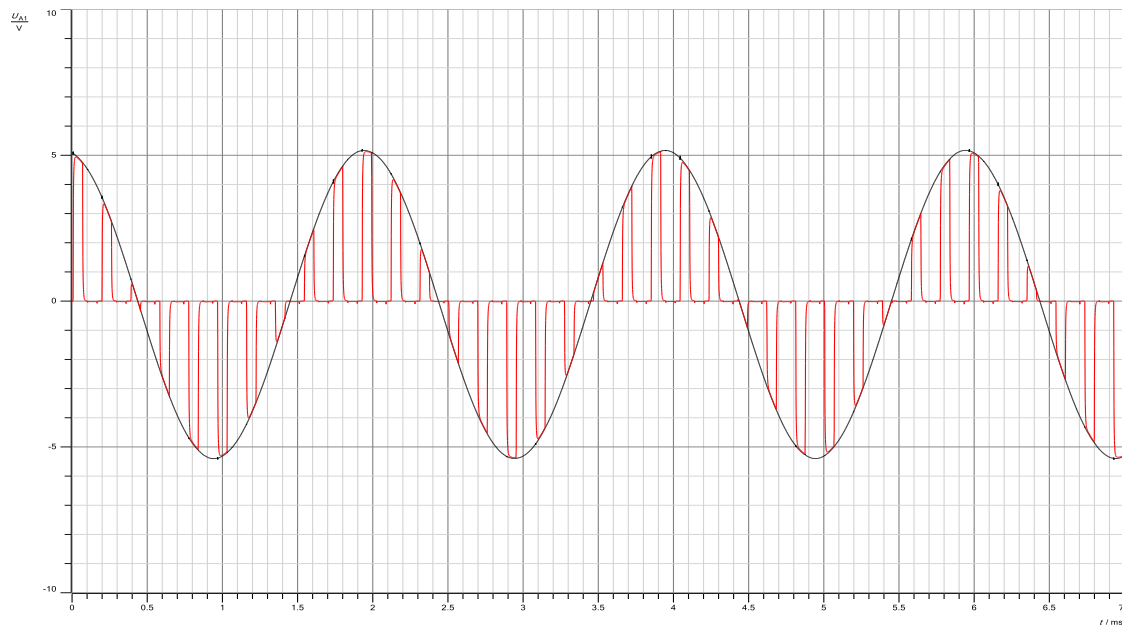


Figure 3.2: the signal at 500Hz & duty cycle 30%

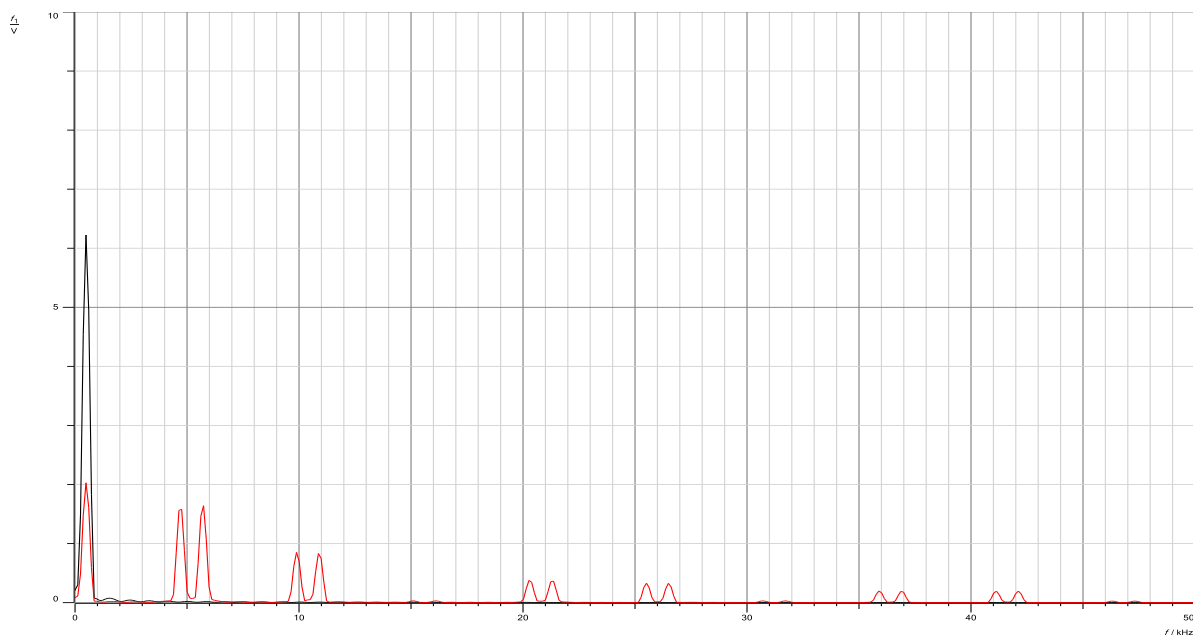


Figure 3.2: the signal at 500Hz & duty cycle 30% in frequency domain

The duty cycle of PAM at 500 Hz impacts the frequency domain, as seen in the images; a lower duty cycle broadens the spectrum, while a higher duty cycle narrows it.

d) **Display the Pulse Amplitude Modulated signal $s(t)$ of PAM2 in the time domain:**

After sitting the required information, the Pulse Amplitude Modulated signal $s(t)$ of PAM2 in the time domain it appeared as in the figure below:

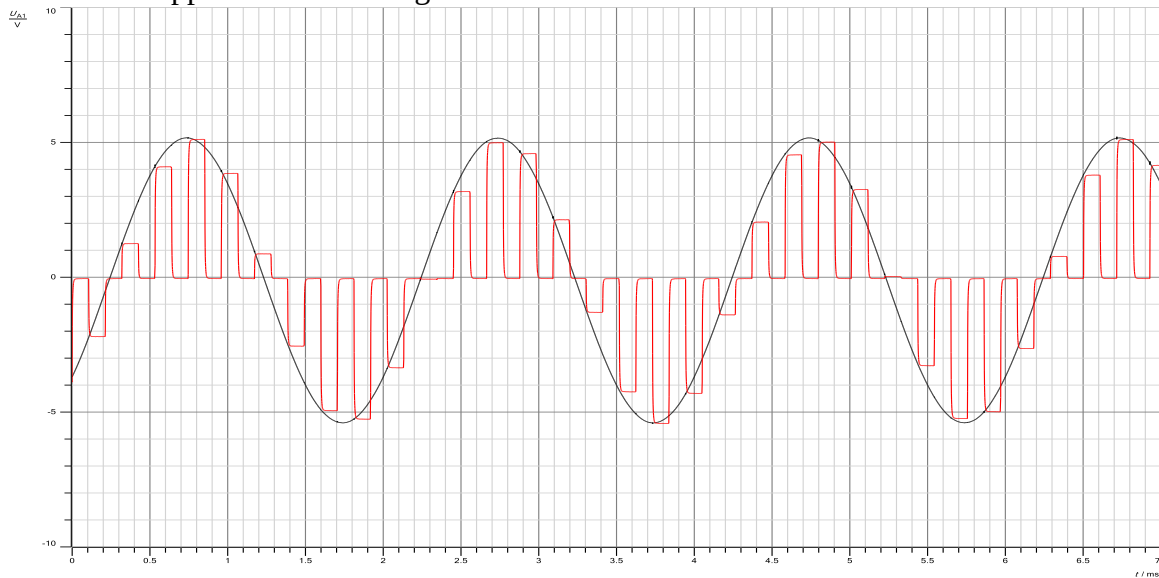


Figure 3.2: PAM signal with flat top pulse

- **Determining the effect of the duty cycle on the PAM in the time domain:**

The duty cycle of the clock generator was reduced to the min value and this is the output:

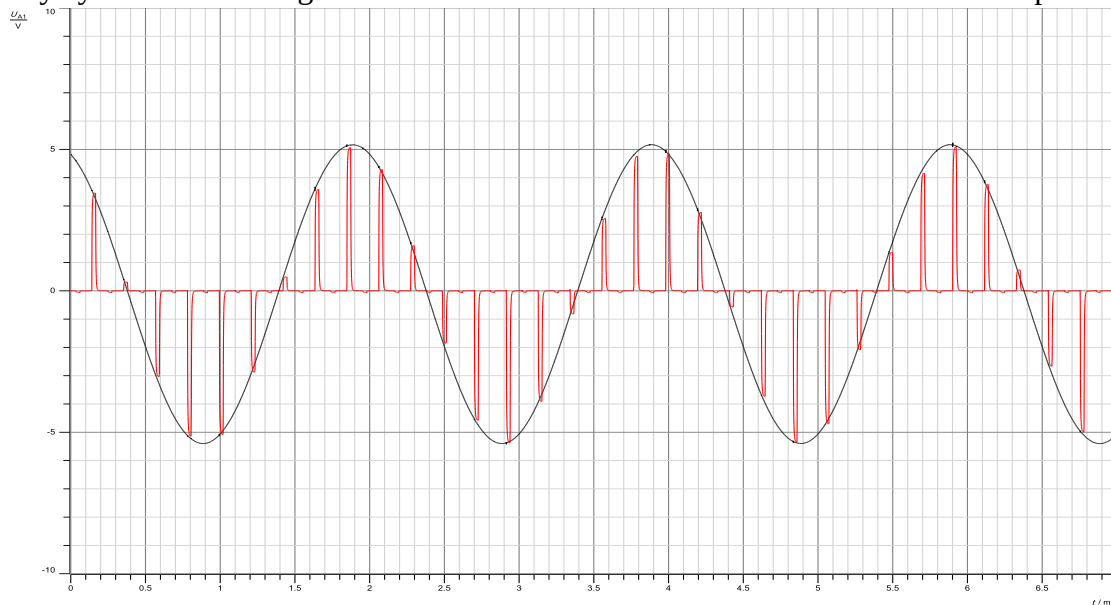


Figure 3.2: PAM signal with flat top pulse at min duty cycle

The duty cycle of PAM with a flat top pulse determines the pulse and transition lengths in the time domain. As mentioned earlier higher duty cycles result in longer pulses and shorter transitions, whereas lower duty cycles result in shorter pulses and longer transitions.

e) **Display the Pulse Amplitude Modulated signal $s(t)$ of PAM2 in the Frequency domain:**

After change the connection as required, this is the output:

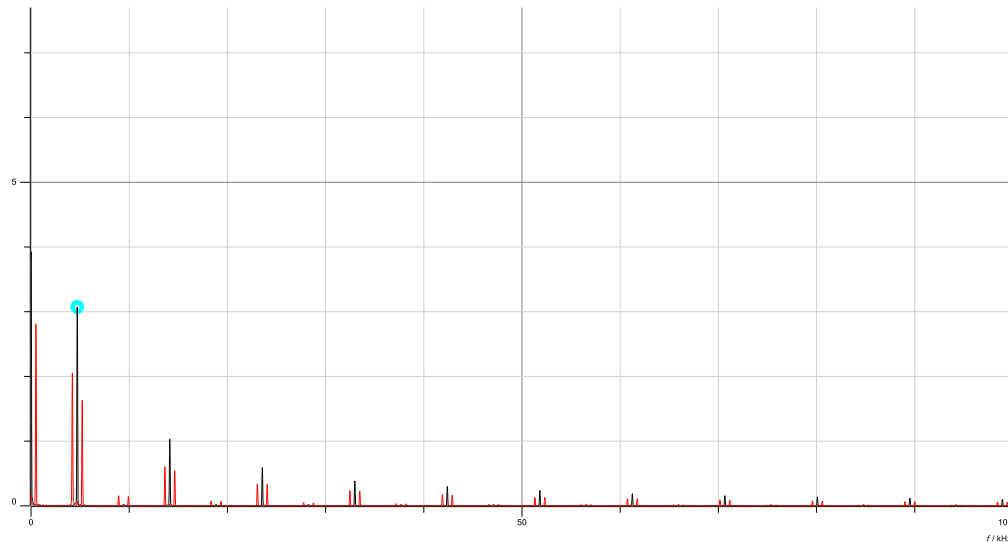


Figure 3.2: PAM signal with flat top pulse in frequency domain

- **Determining the effect of the message frequency on the PAM in the Freq domain:**

Then the message frequency was changed to 1kHz and the output shown as the below:

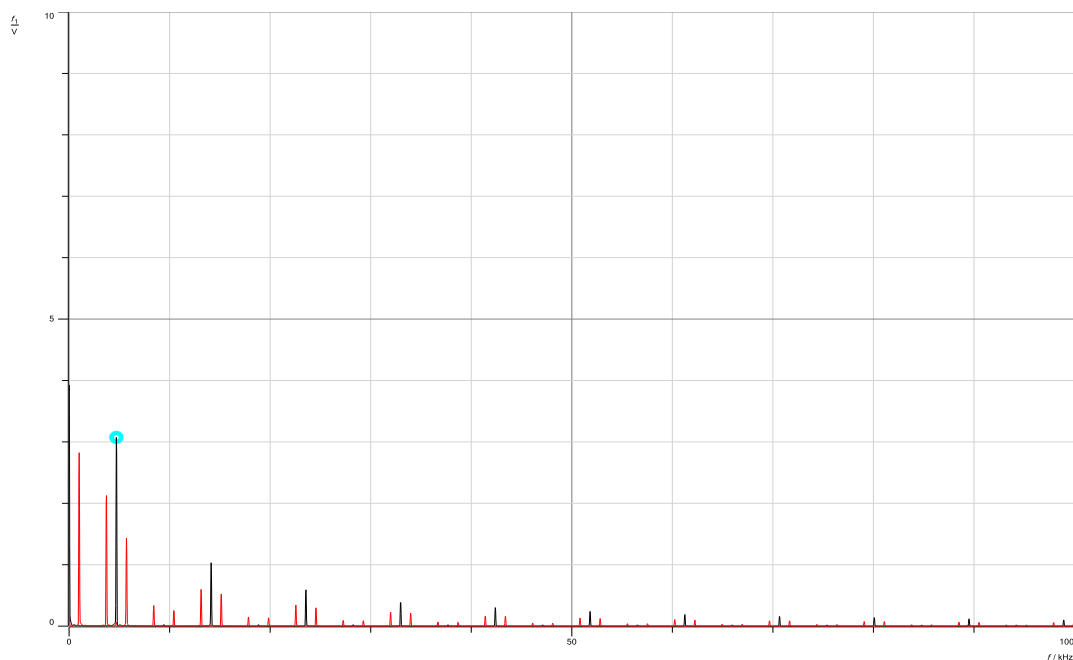


Figure 3.2: PAM signal with flat top pulse in frequency domain at 1kHz

Then at 2kHz:

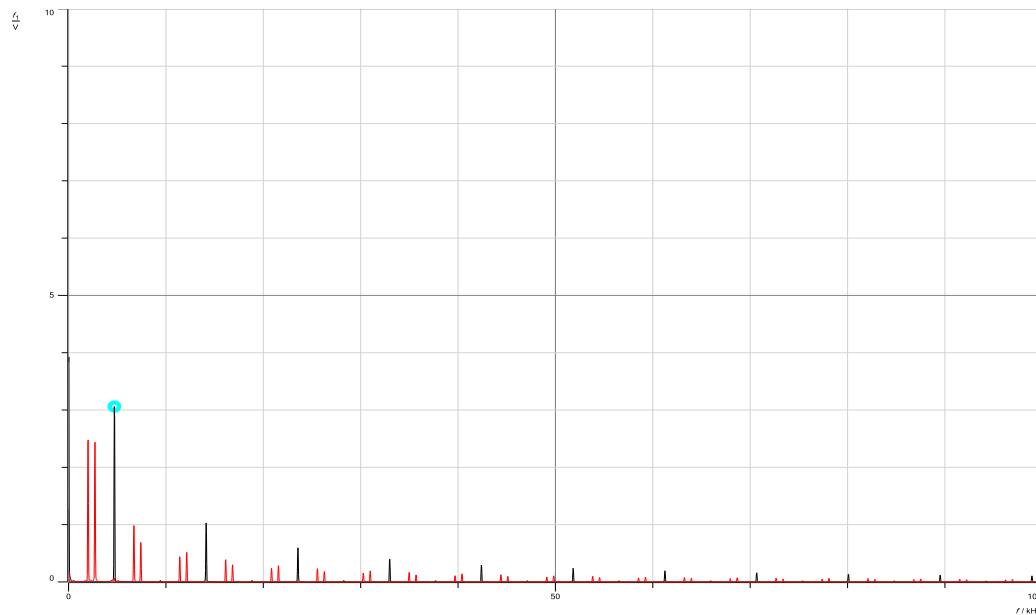


Figure 3.2: PAM signal with flat top pulse in frequency domain at 2kHz

- **Determining the effect of the duty cycle on the PAM in the Freq domain:**

After set the message frequency back to 500Hz, the duty cycle was changed to 10% and this the output:

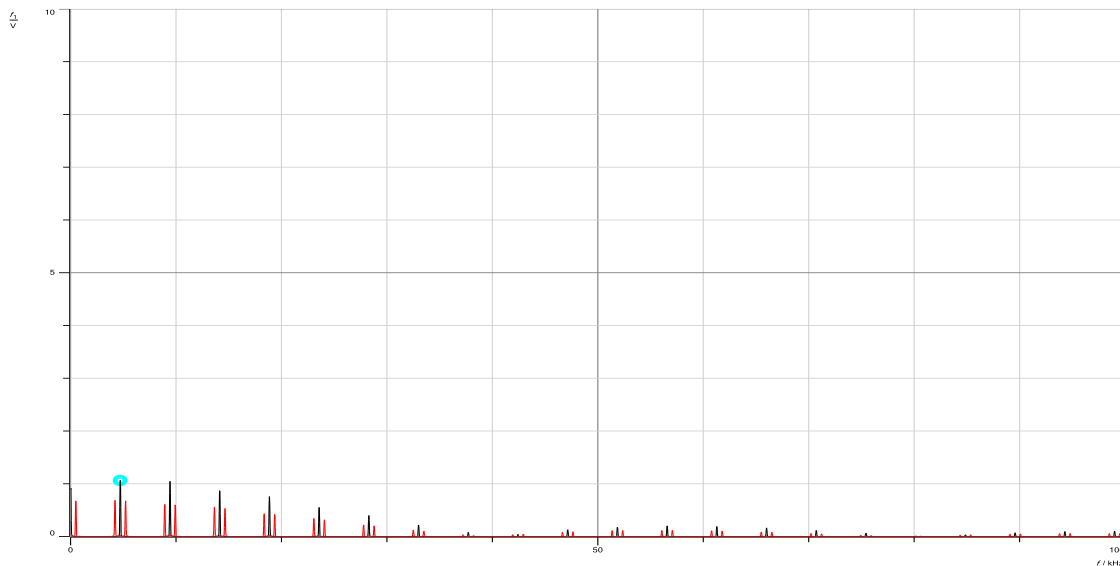


Figure 3.2: PAM signal with flat top pulse in frequency domain at 10% duty cycle

Then at 30%:

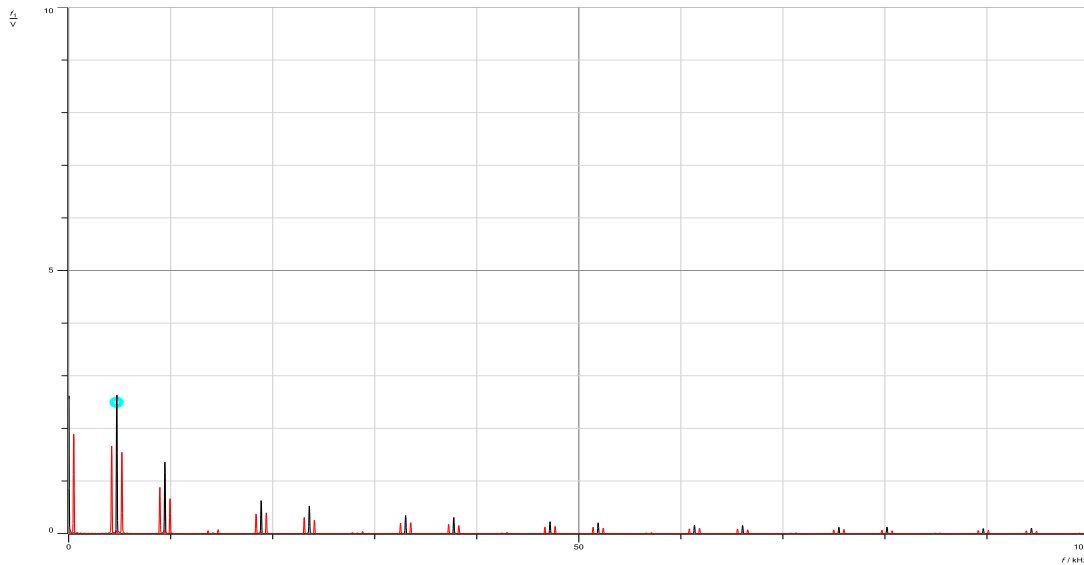


Figure 3.2: PAM signal with flat top pulse in frequency domain at 30% duty cycle

Part 3: Characteristics of Pulse Amplitude Demodulation

a) Display the message signal and the demodulated signal using PAM1:

The duty cycle of the clock generator was set to the max value, and the connection with the required data was set.

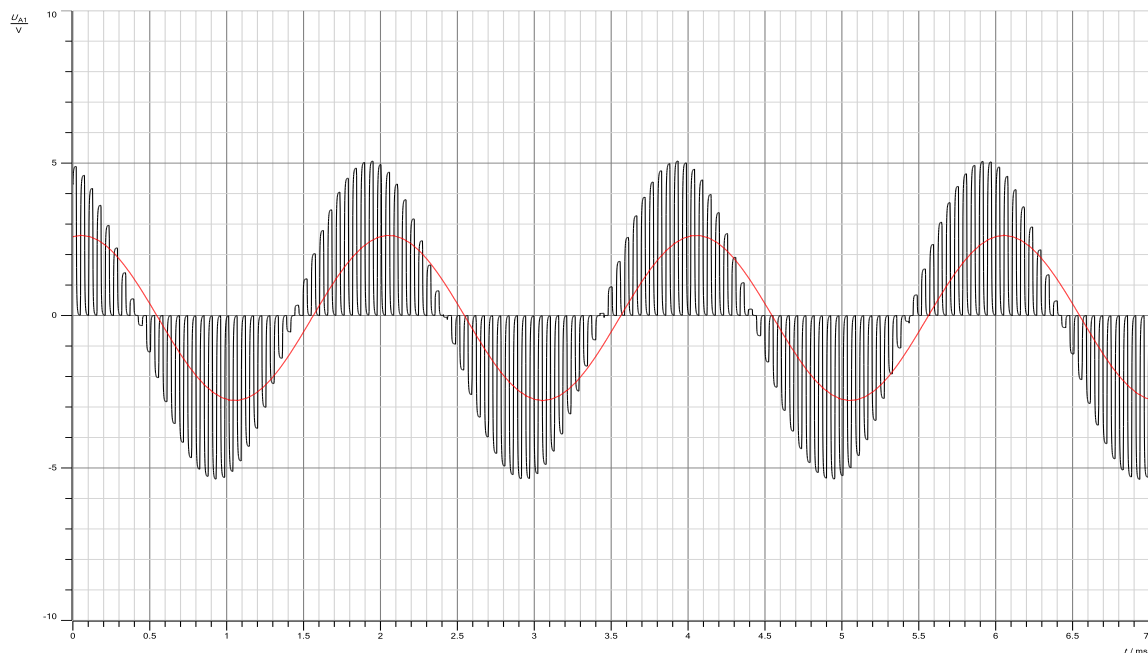


Figure 3.3: Pulse Amplitude Demodulation

b) **Determining the effect of the duty cycle on the PAM in the time domain:**
the duty cycle of the clock generator was changed to 10% and this is the result:

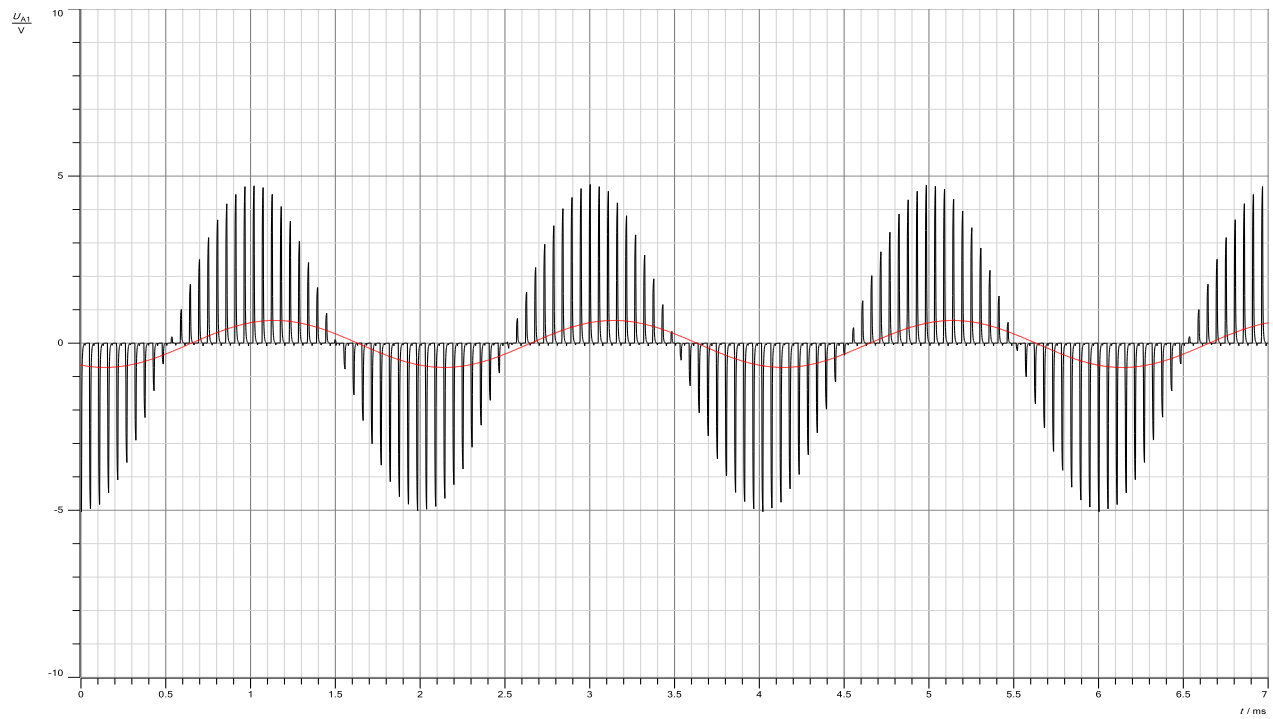


Figure 3.3: Pulse Amplitude Demodulation with duty cycle 10%

Then it was changed to 30%:

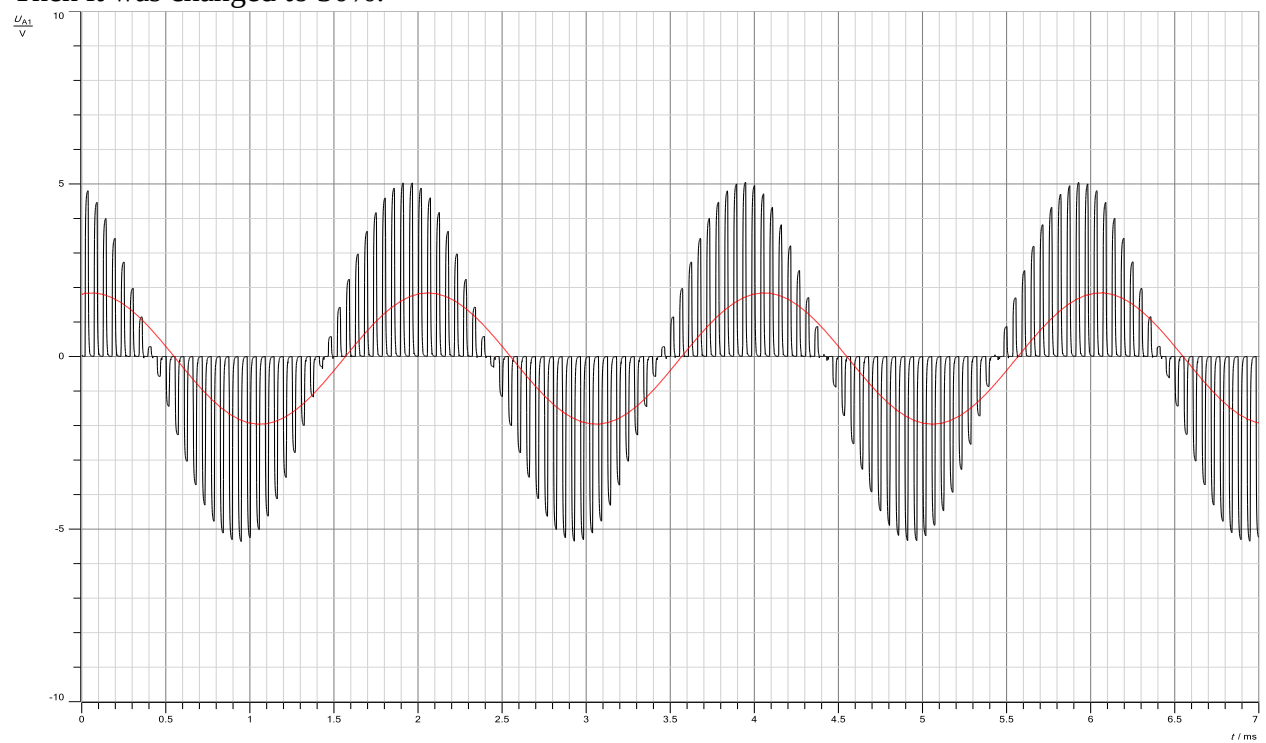


Figure 3.3: Pulse Amplitude Demodulation with duty cycle 30%

Part 4: Aliasing in the Time and the Frequency Domains

- a) **Display the message signal and the demodulated signal using PAM1 in the time domain:**

The connection that required was connected (UB1 to the PAM1 modulator), this the output:

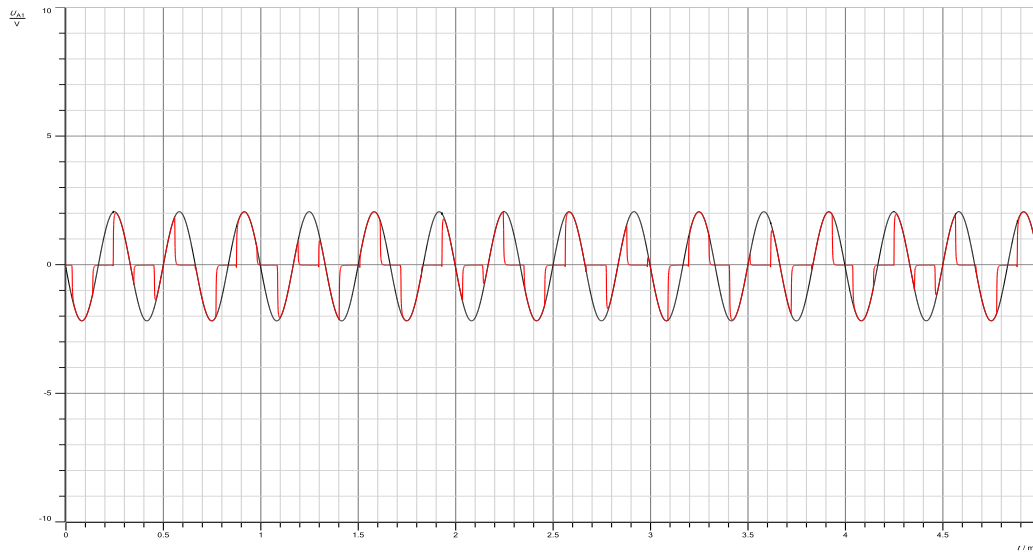


Figure 3.4: the demodulated signal using PAM1 in the time domain

Then connect the Cassy sensor UB1 to the output of the demodulator filter of CH1:

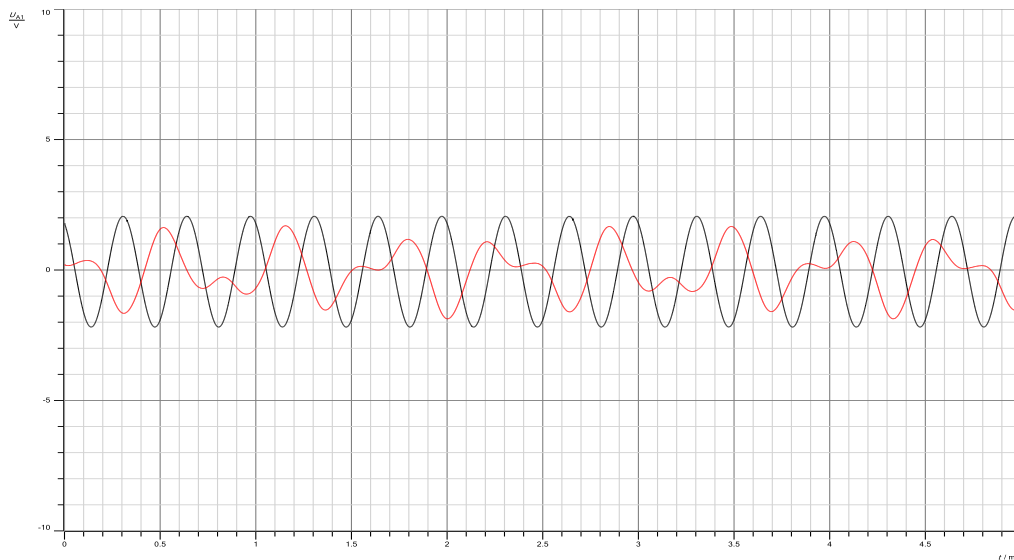


Figure 3.4: the demodulated signal using (demodulator filter) in the time domain

The demodulated signal presented distortion. This is due to a violation of the Nyquist rate rule, since the message frequency exceeds the Nyquist rate.

b) Display the message and the demodulated signals using PAM1 in the frequency domain:

To study the previous part but in the frequency domain, the output of (UB1 to the PAM1 modulator) showed in the figure below:

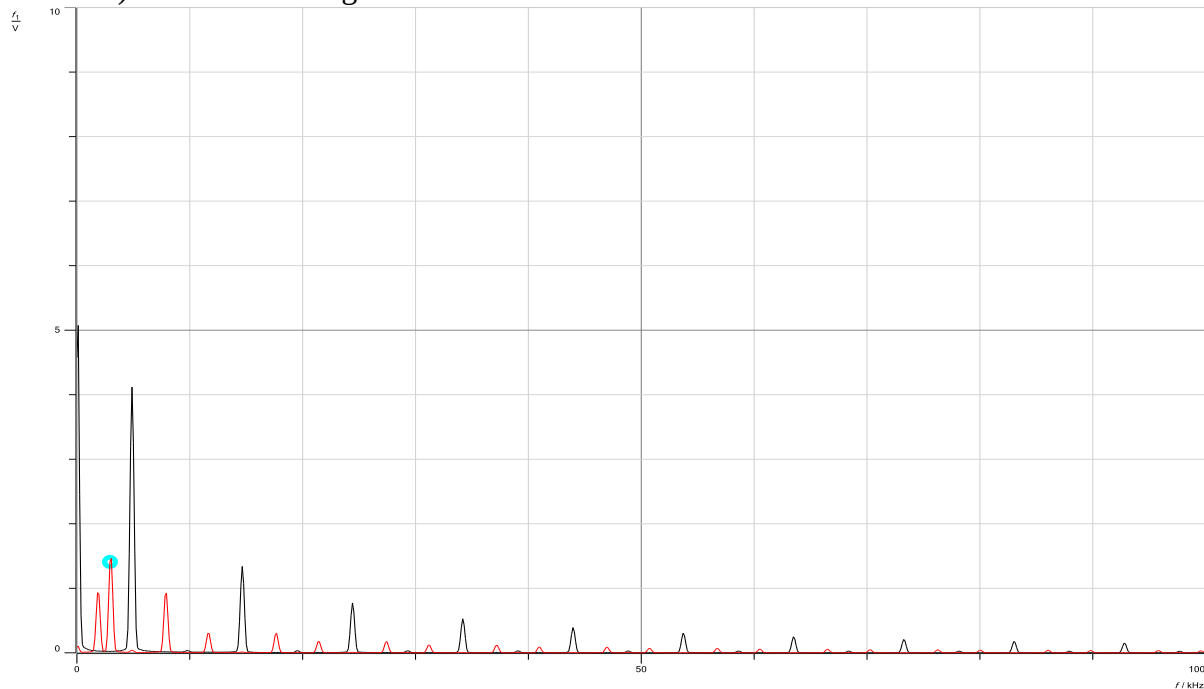


Figure 3.4: the demodulated signal using PAM1 in the frequency domain

Then the Cassy sensor UB1 was connected to the output of the demodulator filter of CH1:

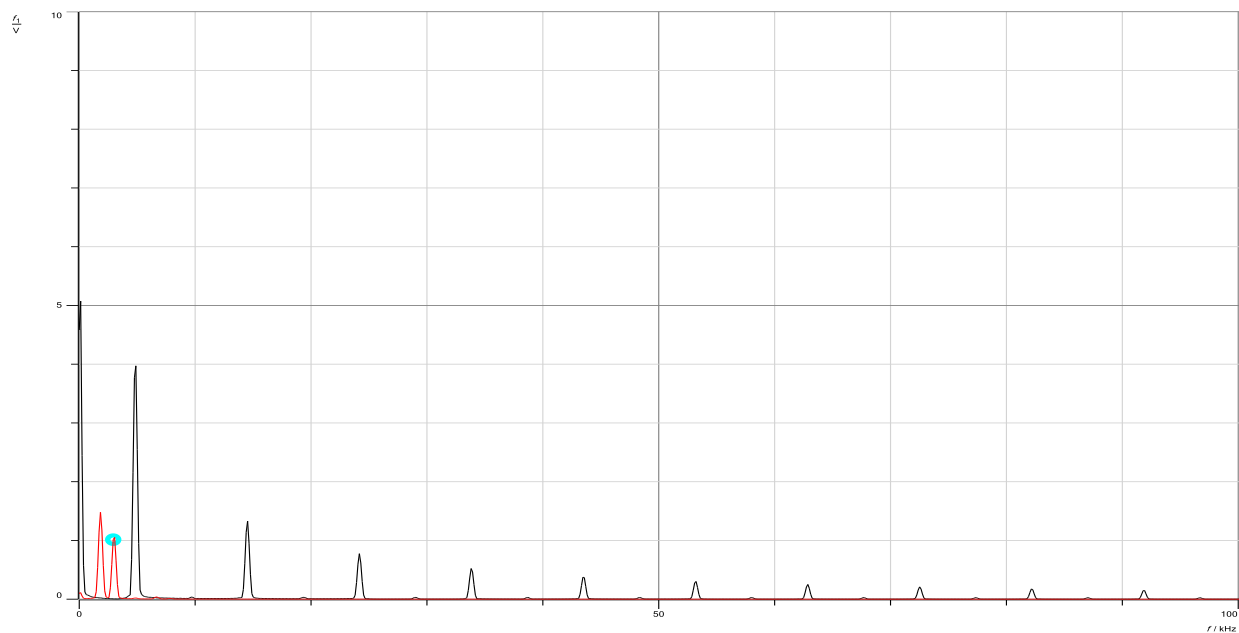


Figure 3.4: the demodulated signal using (demodulator filter) in the frequency domain

Part 5: PAM Time Multiplex

a) Display the time characteristic of the time multiplex signal.

After setting the required information (UA1 to the triangle message channel CH1. UB1 to Output PAM modulator PAM1) this is the output:

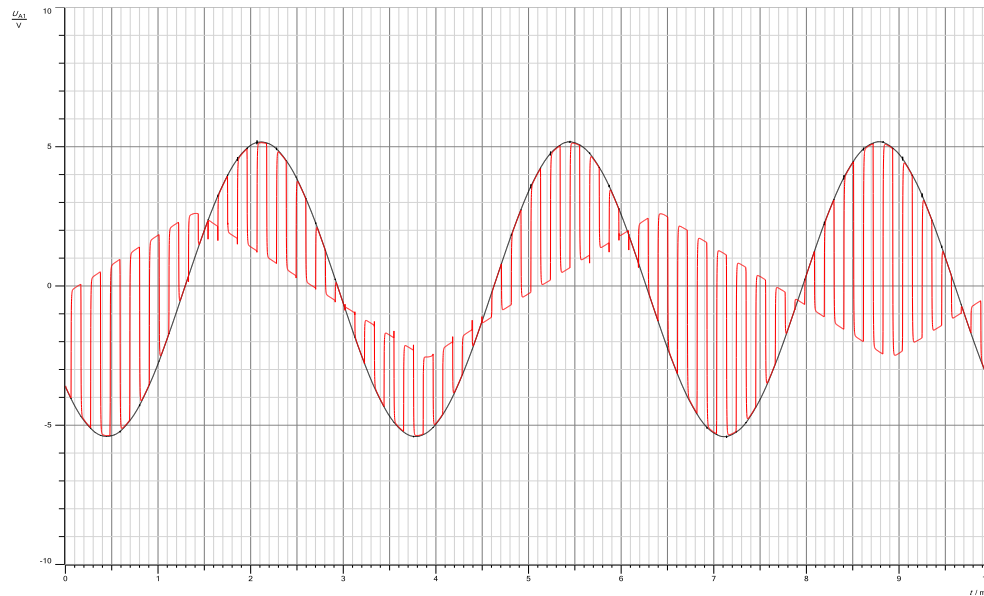


Figure 3.5: the time characteristic of the time multiplex signal (triangle message)

Take note of how the modulated signal follows the triangle wave in certain time slots and the sine wave in others.

Then UA1 was changed to the sine message channel CH1:

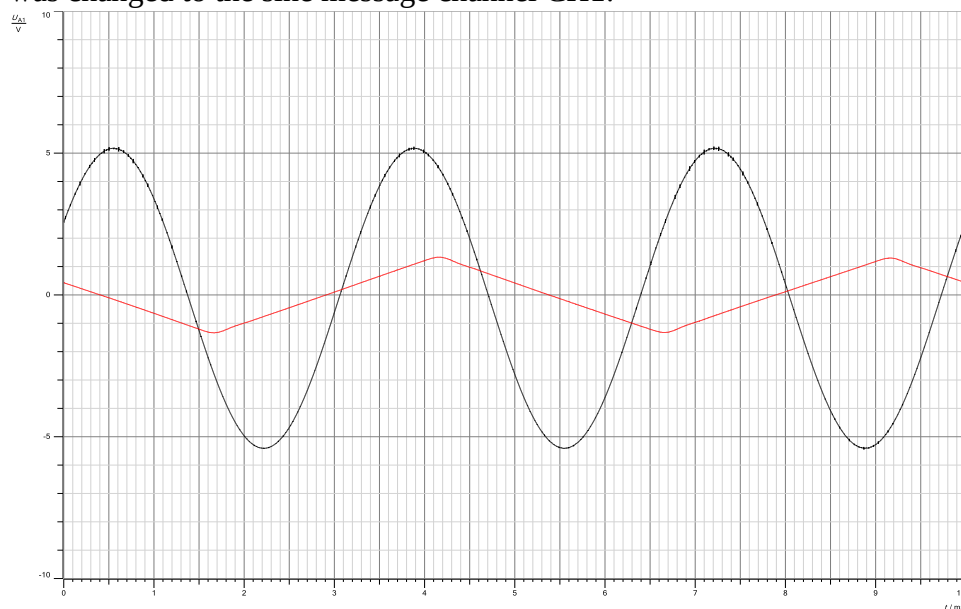


Figure 3.5: the time characteristic of the time multiplex signal (sine message)

And this showed just the triangle wave which it followed.

b) Cross Talk (Δt) left/middle

The effect of cross talk as the phase shift Δt changes and showed below in the time domain:

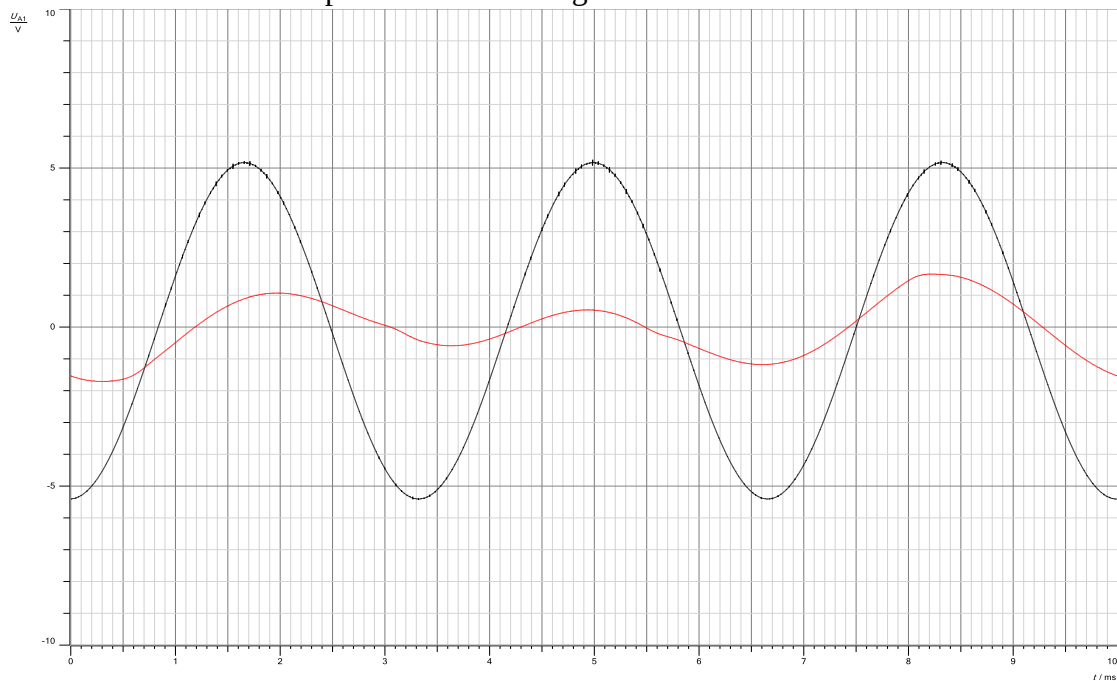


Figure 3.5: the phase shift Δt changes the time domain

The output showed the triangle wave after change the phase shift that led to distortion sine wave.

And in the frequency domain:

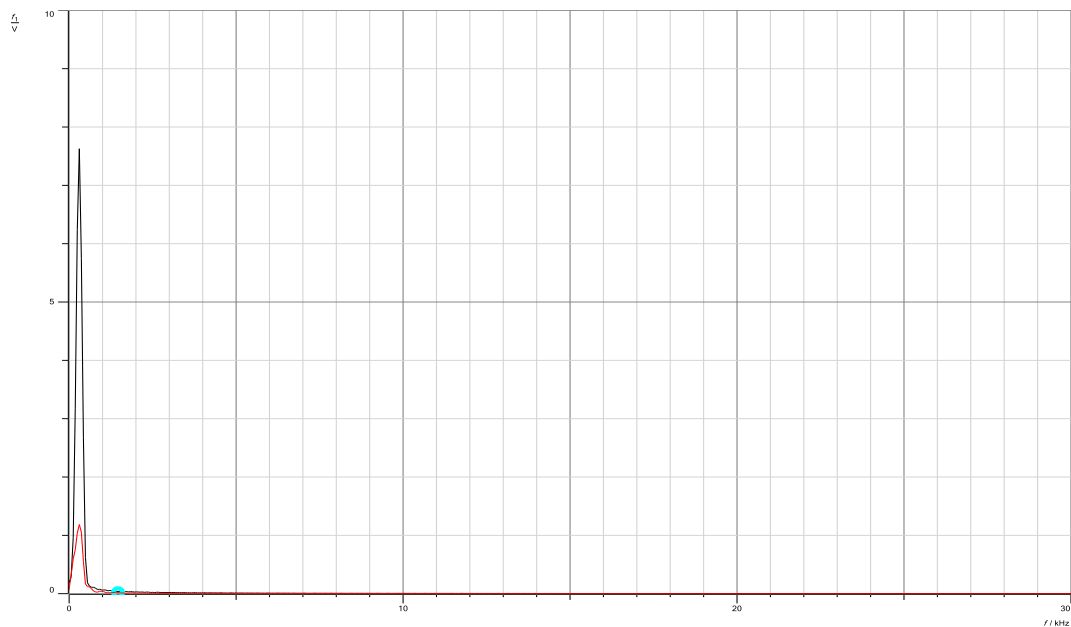


Figure 3.5: the phase shift Δt changes the frequency domain

Then the data was changed to (Triangle, $f_{M1} = 200 \text{ Hz}$, $V_{SS} = 0 \text{ V}$) (Sine, $f_{M2} = 300 \text{ Hz}$, $V_{SS} = 10 \text{ V}$)

- In the time domain:

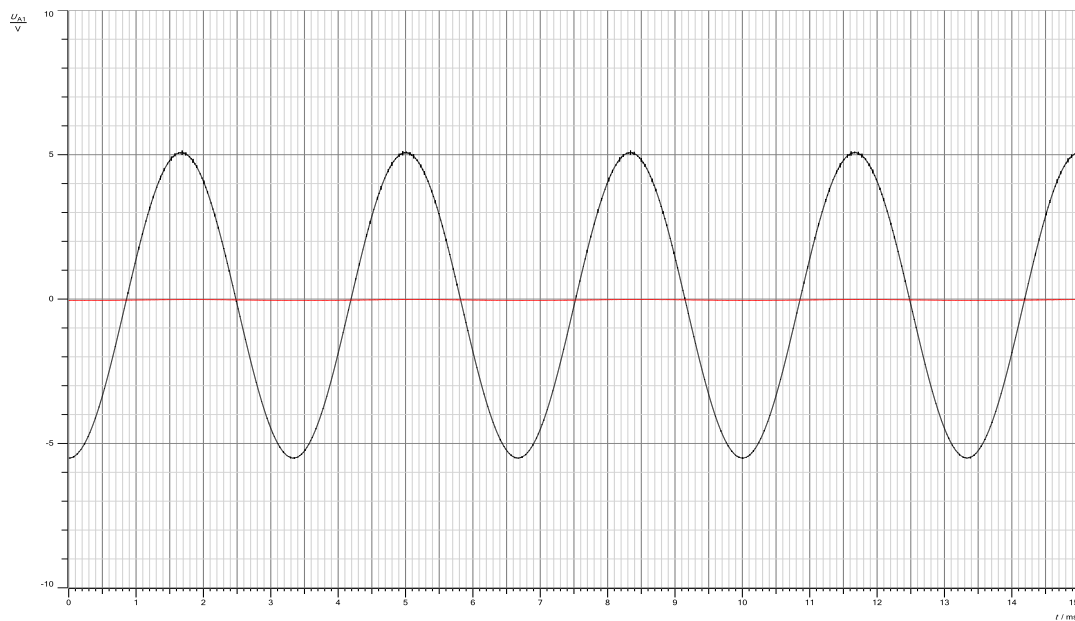


Figure 3.5: the phase shift Δt changes the time domain (triangle & sine)

- In the frequency domain:



Figure 3.5: the phase shift Δt changes the frequency domain (triangle & sine)

The phase shift Δt was changed to until the triangle wave becomes zero.

4.Conclusion

In conclusion, the conducted investigation into Pulse Amplitude Modulation (PAM) provided valuable insights into its temporal and spectral characteristics, demodulation techniques, aliasing phenomena, and temporal multiplexing. Our analysis examined the impact of various parameters, such as message frequency and duty cycle, on the PAM signal in both temporal and spectral domains. By engaging in practical exploration, we gained a more profound understanding of the behavior of PAM and its relevance in modern communication networks. Ultimately, this experiment expanded our knowledge of PAM's significance within current communication technologies.

5.References

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